

Asset-Based Microfinance for Microenterprises: Evidence from Pakistan[†]

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We run a field experiment offering graduated microcredit clients the opportunity to finance a business asset worth four times their usual borrowing limit. We implement this using a hire-purchase contract; our control group is offered a zero interest loan at the usual borrowing limit. We find large, significant, and persistent effects: treated microenterprise owners run larger businesses with higher profits; consequently, household consumption increases, particularly on food and children's education. A dynamic structural model with nonconvex capital adjustment costs rationalizes our results and allows counterfactual analysis; this highlights the potential for welfare improvements through large capital injections that are financially sustainable. (JEL C93, D22, G21, G32, L25, O14, O16)

Is microfinance too “micro”? Can larger financial products generate sustained improvements in microenterprise performance? The first wave of microfinance RCTs found modest average impacts of conventional microcredit contracts on microenterprise performance and practically zero average effects on household consumption (Duflo 2020; Meager 2019). Subsequent work has identified significant heterogeneity in business impacts, particularly among the upper tail of borrowers and those with more business experience (Banerjee et al. 2021; Bryan, Karlan, and

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Osman forthcoming), and several papers show benefits from contractual innovations designed to increase repayment flexibility (Field et al. 2013; Battaglia, Gulesci, and Madestam 2023; Barboni and Agarwal 2018). In their seminal review of the experimental literature, Banerjee, Karlan, and Zinman (2015) recommend that the next generation of microfinance research should explore contractual innovations and noncredit structures, while addressing the lack of evidence for the impact of larger financing amounts on graduated borrowers.

In this paper, we directly address this gap in the literature. We work with one of the most prominent microfinance institutions (MFIs) in Pakistan; that MFI had a large pool of borrowers who had successfully completed previous loan cycles and who wanted to expand their business through the purchase of a fixed asset that cost significantly more than the prevailing borrowing limit. To finance such a large amount in a manner that is satisfactory to the MFI from a risk-reward perspective, we rely on a collateralized asset financing structure that has not previously been used in the experimental microfinance literature: namely, a “hire-purchase” agreement, in which the client’s ownership share in the asset increases as repayments are made. Specifically, we conduct a field experiment in which we offer these graduated microfinance borrowers the opportunity to finance a business asset worth up to approximately US\$2,000,¹ which represents a large capital injection for these clients (approximately four times their previous borrowing limit and substantially more than the loan amounts offered in most of the comparable research). We do this using a hire-purchase contract structure with an 18-month duration, allowing clients to purchase a business asset of their choice; clients are then required to pay rent on the MFI’s proportional ownership share of the asset at the start of each month. Clients who were randomly assigned to our control group were eligible for the MFI’s standard cash loan: a zero interest product with an 18-month duration and a borrowing limit of \$475.

We find a 56 percent average take-up rate of assets for those assigned to any treatment, and low default rates (under 5 percent for both contracts). Most importantly, we find large and significant effects on business and household outcomes, using five rounds of follow-up data in the two years following our intervention. Specifically, treatment clients are more likely to remain in self-employment, have larger businesses (as measured through business assets), better business management practices (particularly in terms of inventory control and purchasing), and greater business performance (on average, an increase in monthly business profits of approximately 11 percent of the control group mean). This generates a significant increase in household income (on average, approximately 9 percent per month) and a significant increase in household monthly consumption expenditure (approximately 6 percent). The bulk of this increased consumption is in household educational expenditure, where we observe a 26 percent average increase compared to the control group. This is predominantly driven by an increase in spending on girls’ education, significant across all measured subcategories: spending on school fees, books and materials, school meals, and transportation costs. We also find significant positive effects on overall purchases of food for the household. Our results are robust to winsorizing

¹ Henceforth, we use \$ to refer to US dollars, based on the actual Pakistani rupee (Re) amounts and the baseline US\$-Re exchange rate of 105.

at multiple levels, to sample-selection concerns (attrition is under 5 percent and uncorrelated with treatment), and to mediation analysis that rules out our results merely being driven by sectoral switching. Our estimates also remain stable when we disaggregate by survey wave.

To understand the mechanisms driving our results—and to consider plausible outcomes under alternative contractual variations—we use a calibrated dynamic structural model of microenterprise capital investment and growth. We build on the structural microfinance approaches developed by Kaboski and Townsend (2011) and Banerjee et al. (2021); we focus on the role of fixed capital and explicitly incorporate the asset-based product that we implemented in our experiment. We find that this model fits the data well—replicating patterns both of estimated treatment effects and a large number of untargated moments—but does so only when we allow for large nonconvex costs of capital adjustment. This implies that nonconvex adjustment costs are crucially important for understanding our estimated treatment effects—in particular, the persistence of our estimated impacts. The model predicts that—as in the data and as in the seminal macroeconomic work of Kaplan and Violante (2014)—households optimally spend down their low-return liquid asset, even though this precludes access to high-return illiquid investments. This framework rationalizes several key features of our data—and, more generally, key features of many microenterprise studies in the literature. Specifically, we observe little or no adjustment to enterprises' fixed capital stock over time, and most households hold minimal wealth in cash or other liquid assets; in our model, this is optimal household behavior notwithstanding that the marginal product of fixed capital in the microenterprise is high. In sum, our model highlights the importance of financial product provision that recognizes lumpiness in investment and the crucial role of large capital purchases for microenterprises. Specifically, the model implies that a microfinance intervention offering a relatively small lump-sum payment will not generate transformational change to the household's circumstances; in contrast, a large transfer can generate sustained improvements in household wealth and income while also being financially sustainable for the MFI.

This conclusion is supported by our analysis of benefit-cost ratios and the internal rate of return (IRR). We show that—under various assumptions about the long-run persistence of treatment effects—our contracts generate very high rates of return. For example, using our estimated treatment effects—and our MFI partner's actual implementation costs—we find a benefit-cost ratio of 3.9 even when assuming zero persistence of effects after the second year of implementation. This rises to a ratio of 8.8 when assuming 5 years of benefits and 11.9 when benefits persist for 10 years. The IRR is 109 percent even when we assume 0 years of persistence, and the IRR converges to 140 percent when assuming 5 years or more of persistence. We then use estimates from our structural model to consider a scenario in which the MFI doubles the rate of interest charged and suffers a doubling of the default rate. Even under this more conservative scenario, we obtain a benefit-cost ratio of 1.7 if we assume zero persistence of effects after the second year of implementation, rising to a ratio of 3.7 when assuming 5 years of benefits and 4.9 when benefits persist for 10 years. The IRR is 9 percent when we assume 0 years of persistence, rising to 31 percent with only 1 year of persistent benefits, 48 percent with 3 years of persistent benefits, and converging to between 53 percent and 56 percent when assuming 5 years or more of persistence.

Our paper contributes to two distinct strands of literature. The first strand has used field experiments to identify the causal effect of microcredit capital injections—often targeted at microentrepreneurs—on business performance and household welfare.² These papers find some evidence of microcredit leading to greater business investment, and indications of gains for upper-tail microenterprises and those with more business experience (Banerjee et al. 2021). Overall, this literature finds modest average impacts on profits and limited evidence of impacts on various measures of household welfare such as consumption; see, for example, the survey by Banerjee, Karlan, and Zinman (2015) and the Bayesian hierarchical analysis by Meager (2019).³ Our paper builds on this evidence, and the recommendations of Banerjee, Karlan, and Zinman (2015), by working with graduated microfinance borrowers looking to purchase a business asset and offering them a much larger financing amount (representing approximately 4 times their previous borrowing limit of \$475). Our financing offer of \$1,900 is significantly larger than the loan amounts offered in most of the existing microcredit literature. One notable exception is the recent work of Bryan, Karlan, and Osman (2022), who focus on the credit allocation decision and the role of psychometric data in predicting the best performers under graduated loans; in their experiment, the control group receives twice the usual loan size, and the treatment group receives four times the usual loan size, with clients given flexibility to determine their loan duration. Our results demonstrate the benefits to business performance and household welfare of “strongly backing” graduated borrowers with a significant relative increase in capital—using a financial contract structure that resulted in the MFI getting its money back, with very few defaults.

The second related strand of literature studies the impact of “big push” asset transfers. Previous work that has provided poor individuals in low-income countries with a large capital injection (usually in the form of productive asset grants) has found substantial and persistent increases in business and household income (see, in particular, De Mel, McKenzie, and Woodruff 2008; Fafchamps et al. 2014; Banerjee et al. 2015; Hussam, Rigol, and Roth 2022; Bandiera et al. 2017; Crépon, El Komi, and Osman 2022; and Balboni et al. 2022). Together, these earlier results beg an important question: Can a big-push microfinance contract generate high investment returns while also recovering the initial capital outlay to be redeployed for future recipients? Our results show that—for graduated borrowers, at least—asset-based financing can provide a sustainable mechanism to generate the high returns identified in the earlier capital-drop studies. In this regard, the only other field evidence we are aware of that shows the impact of asset-collateralized loans is Jack et al. (2023)—who also find very high repayment, with a repossession rate of less than 2 percent (and who, like us, find positive impacts on girls’ education).

² There is also a long tradition of nonexperimental and qualitative approaches to identifying the impact of microcredit, which has produced mixed and sometimes controversial results. For example, see Roodman and Morduch (2014) for a discussion of the earlier work by Pitt and Khandker (1998). For comprehensive surveys of the microcredit literature, see Lensink and Bulte (2019); Cull, Demirgüç-Kunt, and Morduch (2018); and Cai et al. (2021). Bauchet and Morduch (2013) provide an interesting comparison of microcredit and SME borrowers from surveys in Bangladesh. For brevity, we restrict our comparisons here to experimental papers.

³ More specifically, see Augsburg et al. (2015) in Bosnia; Tarozi, Desai, and Johnson (2015) in Ethiopia; Banerjee et al. (2015) in India; Angelucci, Karlan, and Zinman (2015) in Mexico; Attanasio et al. (2015) in Mongolia; Crépon et al. (2015) in Morocco; Karlan and Zinman (2011) in the Philippines; and Fiala (2018) in Uganda.

As in many of the asset-transfer papers, our treatment involves a bundle of related features (indeed, this is true of any microfinance experiment, where each offered product necessarily combines a set of distinct contractual components). For example, Banerjee, Karlan, and Zinman (2015) summarize six microfinance experiments using key contractual features that include (i) client selection, (ii) loan size, (iii) loan duration, (iv) interest rate, (v) repayment frequency, (vi) collateralization, and (vii) individual or group liability. In our setting, we view the key contractual features as being (i) the selection of graduated borrowers as clients,⁴ (ii) the provision of a large loan for the purpose of a business investment, and (iii) same-asset collateralization coupled with repayment according to a hire-purchase schedule. These features are mutually complementary—in the sense that microfinance institutions will typically insist upon long-standing client relationships in order to support substantially larger loan sizes (as in Bryan, Karlan, and Osman 2022) and will similarly require asset collateralization or other securitization methods to manage the increased credit risk (as in Jack et al. 2023; Gertler, Green, and Wolfram 2023; and Carney et al. 2022). Our structural model allows us to consider variations in the key repayment terms (namely, contractual duration and the repayment structure), and these results indicate that our experimental findings are robust to a wide range of plausible contractual variations. Nonetheless, as in the asset-transfer literature (Banerjee et al. 2022), we recognize that there is important scope for future experimental work to test further variations on our basic contractual structure.

Our paper proceeds as follows. In Section I, we summarize our experimental design, and in Section II we report treatment effects. In Section III we present the results from our structural estimation, including counterfactual analysis. Section IV discusses rates of return, and Section V concludes.

I. Experimental Design

A. Study Context

We conducted our study in Pakistan in 2017 and 2018. Microfinance has grown rapidly in the country, with the number of active borrowers more than doubling from 2014 to 2019 and the total loan portfolio increasing by 400 percent over that period (MIX 2019; Ahmed 2019, 2020b). The typical loan in the sector is approximately \$300, to be repaid in 12 months and at annual interest rates ranging from 0 percent to 40 percent (Basharat and Sheikh 2019). Approximately 70 percent of all loans are structured as individual liability. Prior to COVID-19, the sector had maintained very low default rates, with write-offs less than 1 percent of the gross loan portfolio (Ahmed 2019). As of 2019, there were 46 registered microfinance providers in Pakistan, falling into two categories (which, importantly, have quite different funding structures): microfinance banks (MFBs) and nonbank microfinance companies (NBFCs). The key distinction concerns deposits: MFBs are permitted to

⁴Our selection mechanism—namely, our focus on graduated borrowers—is relatively light touch compared to the more sophisticated screening methods attempted in the capital-drop literature; these alternative methods include the use of expert panels, machine learning methods, and methods from mechanism design theory (Fafchamps and Woodruff 2016; McKenzie and Sansone 2019; Hussam, Rigol, and Roth 2022).

accept deposits, whereas NBFCs are not. For this reason, MFBs are regulated by the central bank (whereas NBFCs are regulated by the securities commission). MFBs and NBFCs each serve around half of active borrowers. MFBs' primary source of funding is public deposits, with borrowing constituting less than 10 percent (borrowing is mostly from local banks and development finance institutions). About 75 percent of funds for NBFCs come from debt, provided mainly from the apex funding agency, the Pakistan Microfinance Investment Company, which provides subsidized loans to NBFCs (Malik et al. 2020). In the two years in which we implemented our study, the average NBFC borrowing rate was around 10 percent; in contrast, MFBs paid as much as 15 percent to their depositors (Basharat and Sheikh 2019).

We worked with Akhuwat, a not-for-profit NBFC that provides Islamic microfinance services (though its lending is not restricted to Muslims). As of 2019, Akhuwat was the largest microfinance provider in the whole of Pakistan in terms of both geographical spread as well as number of borrowers—with a market share of around 13 percent, comprising over 891,000 active borrowers across 811 branches, and an outstanding portfolio of Rs16.4 billion (approximately \$106 million at the prevailing market rates) (Ahmed 2020a). Akhuwat receives financial subsidies from the Pakistani government, and its main product is a zero interest loan.⁵ Subsidies are a common feature of the NBFC sector in Pakistan—as it is in many countries (for example, Cull, Demirgüç-Kunt, and Morduch 2018 provide evidence from 1,335 microfinance institutions around the world and show a mean subsidy of 13 percent and a median of 7.6 percent). Similarly, of the 6 prominent microcredit RCTs described by Banerjee, Karlan, and Zinman (2015), 5 of them provided products that were subsidized relative to market rates, and the average subsidy was 15.4 percent in APR terms.⁶

Akhuwat is based in Lahore, the second most populous city in Pakistan and the capital of the province of Punjab. We sampled from microenterprises in and around Lahore. Our sample comprised 757 microenterprise owners who had successfully completed at least one loan cycle with Akhuwat, had reached the maximum permitted borrowing amount (approximately \$475), and had expressed an interest in expanding their business by purchasing a fixed asset. Eligible clients were invited to a workshop, where they completed a comprehensive survey, which included questions asking about individual and household characteristics, household finances, business income, expenditures and assets, and business management practices. Following the survey, all microenterprise owners participated in a set of detailed behavioral games, designed to measure risk preferences, loss aversion, time preferences, and cognitive ability. These are explained in detail in online Appendix Section Q.

⁵ It should be noted that—although Akhuwat's loans are contractually zero interest—clients often make voluntary contributions to the organization. Mahmud and Wahhaj (2019) find that Akhuwat clients donate in the region of 4 percent of their loan amount, and they speculate that this may act as a mechanism for borrowers to signal their quality and obtain larger future loans. Using administrative data for our current sample, we also find evidence of voluntary contributions, in the region of 2 percent of loan amounts. The lower amount in our sample may relate to the fact that our clients had graduated successfully from previous loans and already had access to the maximum borrowing amount.

⁶ The individual differences between implemented rates and market interest rates were 5.3 percent (Bosnia), 12.7 percent (Ethiopia), 35 percent (Mexico), 15.7 percent (Mongolia), and 31.8 percent (Morocco). In India, the implemented product was actually higher than the market rate by 8.1 percentage points.

In online Appendix Table A.2, we compare the characteristics of our sample to a sample of just under 30,000 individuals, covering all of Akhuwat's first-time borrowers in Punjab during the implementation period of our study in 2017 and 2018. The average age in the two samples is very similar (37 and 38 years, respectively). The sample of graduated borrowers has a lower proportion of females (8 percent compared to 41 percent of first-time borrowers). Graduated borrowers also had higher educational attainment; 14 percent had postsecondary, vocational, or university qualifications (compared to 7 percent of first-time borrowers). In contrast, 63 percent of first-time borrowers had a maximum education level of primary school, compared to only 31 percent of graduated borrowers. In terms of sectoral distribution, the most popular sector for first-time borrowers is the service sector, with a proportion higher than in our sample of graduated borrowers (22 percent compared to 7 percent). The second most popular sector among first-time borrowers was retail stores, with a proportion that is again higher than in our sample (21 percent compared to 10 percent). Unsurprisingly, our sample is more heavily tilted toward asset-heavy industries, such as transportation and tailoring (with proportions of 21 percent and 20 percent, respectively, compared to the comparable proportions in the sample of first-time borrowers of 6 percent and 6 percent).

B. Structure of Control and Treatment Contracts

Respondents in our control group were eligible for a zero interest loan over 18 months, up to a limit of \$475 (the MFI's standard upper borrowing limit). Against this, our treatment provided 18-month hire-purchase contracts that allowed clients to finance the purchase of a fixed asset up to the value of Rs200,000 (approximately \$1,900). The contracts charged the equivalent of a 7 percent nominal interest rate and were designed using a sharia-compliant shared ownership structure. The 7 percent nominal interest rate is closer to market rates than Akhuwat's standard product, while still being subsidized compared to average market rates. Nonetheless, the product was designed to break even for Akhuwat given its own cost structure.

Specifically, the contracts obliged clients initially to purchase 10 percent of the asset, with the MFI purchasing the remaining 90 percent. The contracts then require repayments of the MFI's share over the following 18 months. We tested two forms of the contract; these differed in the way that clients were required to purchase the MFI's share. The first version was a *fixed-repayment contract*—in which the client was required to purchase 5 percent of the asset value each month (so that, after 18 months, the client would fully own the asset). The second version was a *flexible-repayment contract*, in which (i) the client was only obliged to purchase 2.5 percent of the MFI's ownership share each month and (ii) the client also had the option to pay *more* than what was required in any given month.⁷ The contract

⁷ If the client purchased all of MFI's share before the 18-month period was over, the contract would terminate. If the client had not fully purchased the MFI's share at the end of 18 months, the contract gives the MFI the right to sell the asset in the market, with proceeds disbursed in proportion to the ownership shares at time of sale. In practice, many clients had repurchased a large share of their asset by the end of the contract, and the MFI decided to allow a few extra months for clients to fully purchase the asset (rather than exercising the sale option), which many successfully did.

structure was that of *diminishing musharakah*, which is a declining-balance agreement that is commonly used to finance the purchase of an asset; it combines two distinct Islamic finance contracts: a shared ownership contract (*musharakah*) and a rental contract (*ijarah*). This type of contract also has strong resonance with Western legal traditions, dating back at least to the ancient Roman law of *hypotheca* (Goebel 1961); in modern legal terms, it resembles a “hire-purchase” contract, which shares features with both “rent-to-own” structures (a more commonly used term in the United States) as well as lease agreements.

Our contracts are based on a “constant amortization” structure rather than “constant payments.” Each month, clients make a fixed payment to increase their ownership share of the asset, as well as a rental payment that is based on the proportional ownership of the asset at the start of the month. The rental amount was based on a nominal annual rate of 12 percent and was chosen (i) to simplify calculations for clients (implying 1 percent of the initial asset value to be paid as rent per month); (ii) to ensure that the MFI would break even in expectation after administrative costs (which were estimated at 7 percent per year, based on historical precedent for the MFI); and (iii) considering the highly subsidized nature of Akhuwat’s funding structure, as discussed in Section IA. Table 1 provides an example of the required payment structure under the fixed-repayment contract for an asset costing \$1,000, where the client has paid \$100 to initially purchase 10 percent of the asset. A nominal annual rental rate of 12 percent implies monthly rent of 1 percent of the asset’s value, which implies a rental payment of \$9 at the end of the first month, reflecting the fact that the MFI initially owns 90 percent of the asset. In addition to the rent, the client is also obliged to purchase 5 percent of the MFI’s ownership share each month, based on the initial asset value of \$1,000, which implies principal payment amount of \$50 per month. At the start of the second month, the MFI’s ownership share is 85 percent, and a reduced rent of \$8.50 is required at the end of the month as well as the regular requirement of \$50 to purchase 5 percent of the MFI’s share. The contract continues in this manner until the eighteenth month, when the client purchases the final 5 percent of the MFI’s ownership share and the contract ends. Over the 18-month duration of the contract, total rental payments are \$85.50 (a raw return of 9.5 percent).

Online Appendix Table A.1 provides two repayment examples for the flexible-repayment contract (again using an initial asset value of \$1,000). The first example illustrates the absolute minimum repayment requirement for the client, which is \$25 per month. Since the MFI’s ownership share decreases more gradually than it does under the fixed-repayment contract, the cumulative rental payments are higher than under the comparable fixed-repayment contract. The second example presents a case where the client repays more than required every month, which results in a more rapidly decreasing ownership share for the MFI (and lower rental payments) and the contract ending at the end of the ninth month.

The procedure for default in both treatment variants is identical: if a client misses a payment, they receive a one-month grace period. If they still do not pay, the asset is repossessed and sold in the market. Proceeds are then disbursed proportional to the ownership shares at the time of the default, reflecting the shared-ownership structure. In practice, we had very few defaults (4 percent of clients); we discuss this further in Section IE.

TABLE 1—CONTRACT REPAYMENT SCHEDULE

Month	% MFI ownership	Payment		Total payment
		Rent	Ownership	
1	90.0	9.00	50.00	59.00
2	85.0	8.50	50.00	58.50
3	80.0	8.00	50.00	58.00
4	75.0	7.50	50.00	57.50
5	70.0	7.00	50.00	57.00
6	65.0	6.50	50.00	56.50
7	60.0	6.00	50.00	56.00
8	55.0	5.50	50.00	55.50
9	50.0	5.00	50.00	55.00
10	45.0	4.50	50.00	54.50
11	40.0	4.00	50.00	54.00
12	35.0	3.50	50.00	53.50
13	30.0	3.00	50.00	53.00
14	25.0	2.50	50.00	52.50
15	20.0	2.00	50.00	52.00
16	15.0	1.50	50.00	51.50
17	10.0	1.00	50.00	51.00
18	5.0	0.50	50.00	50.50
Total		85.50	900.00	985.50

Notes: This table provides an example of the required payment structure under the fixed-repayment contract for an asset costing \$1,000, where the client has paid \$100 to initially purchase 10 percent of the asset. The table shows monthly rent payments of 1 percent of the asset's value; in addition, the client is also obliged to purchase 5 percent of the MFI's ownership share each month. Over the 18-month duration of the contract, total rental payments are \$85.50 (a raw return of 9.5 percent).

C. Descriptive Statistics

Online Appendix Section D presents summary statistics for the 757 microenterprise owners: 92 percent were male, with an average age of 38 and 7.5 years of formal education; 84 percent were married, and the average household size was 6, of which 2 people were typically earning some form of income. Average monthly household income was \$353 (median \$295), and average monthly household consumption expenditure was \$211 (median \$180), which puts our average household in the second quintile of the overall distribution for household consumption in Pakistan (Pakistan Bureau of Statistics 2017). The mean number of businesses in the household was 1.2, and the average microenterprise owner had 9.6 years of experience in their current business. The mean number of employees was 0.9, with a median of 0. The most popular business sectors were (i) transportation, primarily involving rickshaws as well as other transportation assets (and comprising 21 percent of the sample); (ii) tailoring and textile-related trades, including sewing of footwear and other fabric- and garment-related activities (20 percent of the sample); (iii) various forms of manufacturing and related trades (11 percent of the sample); (iv) food and drink businesses (10 percent of the sample); (v) various types of retail shops and market traders (10 percent of the sample); (vi) construction and related trades (9 percent of the sample); (vii) professional services, including telecommunications-related services (7 percent of the sample); and (viii) photography and other entertainment-related sectors (6 percent of the sample).

Average monthly baseline business profits in our sample were \$245 (median \$219),⁸ and the average value of total fixed assets for the business was \$920 (median \$361).⁹ This shows that the financing amount offered to our treatment group could triple the stock of fixed assets for the average firm (and was five times the median firm's fixed asset stock). In comparison, of the six microcredit field experiments summarized in Banerjee, Karlan, and Zinman (2015), five were targeted at microenterprises. In those five, the mean loan size offered in US dollar PPP terms was \$909, with a median across the studies of \$696. Our product therefore represents a sizable capital injection relative to most of the literature and importantly, provides a large multiple of the prevailing borrowing limit for the microenterprises in our sample.

D. Treatment Assignment, Take-Up, and Assets Chosen

Assignment Mechanism.—Participants were randomly assigned into one of three groups: (i) the control group; (ii) treatment group 1, who were offered the fixed-repayment hire-purchase contract to purchase an asset up to the value of \$1,900 (and if they rejected the offer, they were also eligible for the \$475 zero interest loan like the control group); and (iii) treatment group 2, who were offered the flexible-repayment hire-purchase contract to buy an asset up to the value of \$1,900 but were free to reject the offer of flexibility and take the fixed-repayment contract (and were also free to reject both contracts and take the \$475 zero interest loan). In this section, we describe the treatment assignment procedure and overall take-up patterns.

We assigned respondents to treatment using matched sextuplets (Athey and Imbens 2017), where we stratified on gender, microenterprise business type, and profits. We describe this process in online Appendix Section B. Online Appendix Table A.3 reports normalized differences between our control group and our treatment groups (as recommended by Imbens and Rubin 2015) and show that our sample was well balanced.

During the baseline workshop, after participants had completed their surveys and behavioral games—but before any randomization of contracts had taken place—the fixed-repayment contract was described to everyone, using a vignette and example calculations (see online Appendix Section R for details of the script). The flexible-repayment contract was not demonstrated at this stage; rather, we preferred to introduce clients gradually to the calculations for principal and rent using the simpler-to-understand fixed-repayment contract, which we later used as a reference point

⁸ Average monthly revenues were \$728, implying what appears to be a high profit margin of approximately one-third. However, our measure of profits is defined as “net income after all expenses but *before* paying one's own wage,” which would bring the true profit margin down to more conventional levels (another difference with the standard accounting measure of profit is that we do not account for depreciation expenses, which would be subtracted from net income in a conventional income statement).

⁹ The value of business assets was based on the response to the question “How much it would cost you to replace the assets with ones in similar condition?” In all baseline and follow-up surveys (for both treatment and control), enumerators took photos of both fixed and current assets, to improve reporting accuracy. In our definition of business fixed assets, we excluded buildings and land, which are notoriously difficult to value and which were not permitted as a purchase in our project. We also measured current assets, which on average consisted of \$317 of inventory, \$127 accounts receivable, and \$188 business cash reserves. Note that the definition of business fixed assets requires ownership; at baseline, we find little expenditure on rented machinery: an average of \$10 per month, with zero spending up to the seventy-fifth percentile. This rules out the possibility that our treatments are just shifting people from renting assets to owning them (and there does not appear to be a large rental market that provides access to the kind of fixed assets that microenterprises in our sample demanded).

when explaining the flexible-repayment contract to a randomly selected subgroup. At the end of the workshop, all participants were given a one-page information sheet and allowed a few days to consider the product (the fixed-repayment contract).

Participants were subsequently visited by MFI field officers and research assistants, who were given a tablet computer with a preprogrammed survey form that contained the treatment status of all participants. Field officers were not informed of the treatment status of the client they were visiting. Individuals randomized into the control group were informed that they would not be offered the contract but that they would still be eligible for the zero interest loan of \$475 from the MFI. Individuals who were randomized into and accepted the fixed-repayment contract began the contract signing and asset procurement process with the MFI.

A third group were randomly selected to be offered the flexible-repayment contract, while being given the opportunity to reject the offer and still take the fixed-repayment contract. The flexible contract was explained to them as being similar to the fixed-repayment contract but with the added optionality that they would only be required to make a 2.5 percent ownership payment every month, compared to the required 5 percent monthly ownership payment for the fixed-repayment contract (which nests the flexible-repayment contract). All other aspects of the contract were identical. Individuals were then given a one-page document with a simple summary of the structure of the flexible-repayment contract, with diagrams and tables to illustrate the repayment schedule. Participants were informed that they would be visited after a few days to take their decision on whether they would accept the flexible-repayment contract, with contract signing (for whichever of the two contracts they chose to accept, if any) and asset procurement taking place shortly after. As such, we used the same in-person visit protocol, decision elicitation procedure, and “cooling-off period” as for the fixed-repayment contract.

When describing the products to participants, we were careful not to use Arabic terms nor any other words that might carry religious connotations; instead, we used the local Urdu terms for joint ownership (*shirakat* rather than the Arabic *musharakah*) and rent (*kirayah* rather than the Arabic *ijarah*). We took this approach in order to avoid potential complications that might otherwise arise from religious-moral incentives to repay (an issue explored in detail in a consumer finance context by Bursztyn et al. 2019).¹⁰

Contract Take-Up.—We assigned 254 microenterprise owners to control. Of the 257 individuals assigned to the fixed-repayment contract, 53 percent accepted the offer, successfully provided the required 10 percent initial payment, and proceeded with contract completion and asset procurement. Of the 246 individuals assigned to the flexible-repayment contract, 50 percent accepted it and proceeded to contract completion and asset disbursement, 9 percent rejected the flexible-repayment contract but

¹⁰Bursztyn et al. (2019) explore the role of morality in debt repayment using an experiment with an Islamic bank in Indonesia; the authors work with a sample of customers who had missed their repayment date and send them various text messages to encourage repayment. In their main treatment, the authors find that a religiously framed message—highlighting that failure to repay (when one is able to do so) violates a moral norm—significantly increases debt repayment. Through several follow-up experiments, the authors conclude that it is actually the moral statement—rather than the religious language and/or the use of Arabic words—that drives the main effect.

did take the fixed-repayment contract and ended up with the asset, and 41 percent took neither contract.¹¹

In total, 281 participants accepted one of the two treatment contracts (157 under the fixed-repayment schedule and 124 under the flexible-repayment contract), provided their 10 percent initial payment, and had their asset purchase financed. Under both the fixed and flexible contracts, microenterprise owners were permitted to purchase a business fixed asset of their choice worth up to Rs200,000 (\$1,900). The client (not the MFI) was responsible for selecting the particular asset and the asset supplier. Further, the MFI was not responsible for assisting the microenterprise owner in using their asset or in its maintenance. The role of the MFI (after some light screening to ensure that it was a self-contained fixed business asset) was simply to provide financing for 90 percent of the value of the asset and to collect payments.

The mean asset value was \$1,517 (median \$1,666), and approximately one-third of clients chose the maximum financing amount possible. The maximum treatment financing amount available was decided in advance by the MFI, based on their risk appetite and their assessment of typical fixed asset prices. The MFI did allow respondents to purchase up to three assets, provided that the assets formed a complementary bundle. Individuals who chose the maximum financing amount possible were more likely to choose a bundle of assets rather than a single asset.¹² Approximately another one-third of clients purchased assets costing between \$1,500 and \$1,900, with the remaining third purchasing assets worth between \$500 and \$1,500 (with a spike at \$1,000). A closer analysis reveals the importance of indivisible investments across a number of sectors. It is *not* the case, for example, that the upper mode of the distribution is driven by one or two of the most popular asset categories; instead, we see a high average purchase price across almost all asset categories. For example, for the most popular asset categories, mean purchase prices

¹¹ We collected information on reasons for refusal among those who refused at the time of the contract offer. However, most of those who were assigned to treatment but did not take up had actually notionally accepted the treatment when offered, but they then declined by being unwilling or unable to produce the necessary deposit. We did not directly collect information on reasons for refusal for these respondents (though, of course, one can reasonably presume that being unwilling or unable to produce the necessary deposit would be a primary reason for many of these respondents; this is also consistent with the results in Table 3 that we discuss shortly and which show that individuals who took the contract were wealthier than those assigned to treatment but who did not take up). Nonetheless, we do have some data from loan officers, some of whom collected information on reasons for clients not taking up at the point that the loan officers tried to collect the deposit and sign the final contract. Specifically, we have these data for 40 percent of the 222 respondents who were assigned to treatment but did not take up the contract. Often, the reasoning given is not much more than them stating that they decided not to continue. Some of the common (nongeneric) reasons include (i) an inability to gather the funds for the 10 percent initial payment required; (ii) a decision, after considering the required monthly payments, that it was too high for them; (iii) a decision that what they needed for their business was actually working capital or a combination of working capital and fixed assets (which we were not providing in this project, which was limited to fixed assets).

¹² One way to see this is by analyzing the most popular asset, a rickshaw. Only 4 percent of those who chose the maximum financing amount purchased a standalone rickshaw, while 47 percent of the sample who didn't "max out" the financing amount chose a rickshaw. More generally, typical combinations of assets include (i) in the tailoring and textile-related sector, a *Juki* machine (which produces specialized stitches for a variety of items, including clothing, shoes, and bags), an overlock machine (which sews over the edge of one or two pieces of cloth for edging, hemming, or seaming and is also used for inserting zippers), and a picot machine, which does more decorative types of stitches; (ii) in the food and drink sector, cookers/ovens combined with food counters; (iii) in manufacturing, a combination of welding tools, metal cutting machines, and a lathe machine for shaping metal and wood; (iv) a combination of computer-related items, including laptops, printers, photocopying machines, and scanners. The MFI required that each funded asset should be a standalone object; for example, a small number of clients were refused permission to purchase building materials, which would have been incorporated into a larger structure (and thus almost impossible to repossess in case of default).

were (i) \$1,626 (rickshaws), (ii) \$1,504 (sewing machines), (iii) \$1,621 (cameras), (iv) \$1,283 (manufacturing/welding machines), (v) \$1,626 (lathe machines), (vi) \$1,476 (food machines). Online Appendix Figure A.2 illustrates the distribution in the value of assets financed for clients who took up one of our treatment contracts. Online Appendix Section F presents results from regressions that investigate the relationship between contract assignment and the value and type of asset chosen by microenterprise owners. The average value of asset financed for those assigned to the fixed-repayment contract was \$1,471, while those assigned to the flexible-repayment contract chose assets with an average value of \$1,530.¹³

In Table 2, we compare the characteristics of individuals who took up an asset finance contract with those who were assigned to either of the treatment contracts but did not take up the product. There is no significant difference in average age or gender, nor is there any difference in terms of cognitive ability (measured using a series of mathematical questions) or from an index of business management practices. There is also no difference in the proportion who work in the two most popular sectors, transportation and tailoring (which jointly account for 40 percent of our sample). We *do* find that contract takers have larger and more profitable businesses. On average, they have 6 percent higher revenues, 18 percent higher profits, 38 percent greater fixed assets, and 34 percent higher business cash holdings. They also have 80 percent greater household savings and 12 percent greater monthly consumption expenditure.¹⁴

In Table 3, we report on total borrowing for all individuals in the sample within the first three months of the experiment, including both cash- and asset-based loans. In short, the intervention significantly increases total borrowing. Specifically, column 1 shows that assignment to treatment (being offered the asset-based loan, pooling both the fixed- and flexible-repayment contract offers) led to a 48 percentage point increase in the likelihood of any borrowing from Akhuwat (compared to a control mean of 13 percent). Columns 3 and 5, respectively, show that this is composed of an 8 percentage point decrease in the likelihood of taking a cash loan from Akhuwat (compared to a control mean of 13 percent taking cash loans) and a 56 percentage point increase in the likelihood of taking an asset-based loan (compared to a control mean of 0). Column 10 shows that the amount of extra borrowing in the treatment group is very large: \$821 greater borrowing from Akhuwat, compared to a control mean of only \$40.¹⁵

¹³The difference in means is not significant when controlling for stratification dummies in a regression ($p = 0.233$). Column 2 of the unnumbered Table in online Appendix Section F provides some suggestive evidence of more risk-averse individuals choosing higher asset values when offered the flexible contract. The remaining columns show that, for the five most popular assets, there is no clear difference by treatment assignment in the proportion of microenterprise owners choosing that asset.

¹⁴In online Appendix Table A.39, we explore heterogeneous take-up separately for each contract using some prespecified behavioral characteristics.

¹⁵In Table 3, we define the short run as any loan taken within three months of participants entering the experiment. In online Appendix Section E, we provide a more detailed breakdown of cash borrowing from Akhuwat administrative data (panel A of online Appendix Table A.4) as well as cash borrowing from all sources as obtained from the survey data (panel B of online Appendix Table A.4), at different time periods. Column 1 of panel A of online Appendix Table A.4 shows that, 1 month after the start of the project, 5 percent of the control group had taken a cash loan from Akhuwat. Columns 2 to 4 show that this increases to 13 percent by the 3-month stage (which is the figure that corresponds to Table 3), 17 percent by the 6-month mark, and 31 percent by the 18-month mark (which is the duration of the asset finance contract). In terms of the dollar amount of cash borrowing (again, using administrative data from Akhuwat), at the 6-month mark, the average for the control group was \$53.88. This can be compared

TABLE 2—CHARACTERISTICS OF TAKE-UP GROUP

Variable	Take-up = 0 Mean/SE (1)	Take-up = 1 Mean/SE (2)	Total Mean/SE (3)	t-test difference (1) – (2)	Normalized difference (1) – (2)
Age	38.81 (0.64)	37.90 (0.63)	38.30 (0.45)	0.91	0.09
Female	0.12 (0.02)	0.08 (0.02)	0.10 (0.01)	0.04	0.12
Math score (above median)	0.55 (0.03)	0.53 (0.03)	0.53 (0.02)	0.02	0.04
Management practices (above median)	0.50 (0.03)	0.51 (0.03)	0.50 (0.02)	–0.01	–0.03
Sector: transportation	0.21 (0.03)	0.20 (0.02)	0.20 (0.02)	0.01	0.03
Sector: tailoring	0.19 (0.03)	0.21 (0.02)	0.20 (0.02)	–0.02	–0.06
Business revenue	706.09 (49.47)	744.87 (46.65)	727.75 (33.97)	–38.78	–0.05
Business profits	227.65 (9.80)	269.11 (10.17)	250.82 (7.19)	–41.46	–0.26
Total fixed assets	789.16 (98.33)	1,086.29 (103.36)	955.15 (72.46)	–297.13	–0.18
Current assets: Cash	163.65 (18.09)	219.81 (17.72)	195.02 (12.77)	–56.16	–0.20
Household savings	308.61 (49.07)	557.42 (58.30)	447.61 (39.46)	–248.80	–0.28
Household loans	55.98 (6.30)	25.44 (3.50)	38.92 (3.46)	30.54	0.39
Household consumption expenditure	203.13 (7.62)	227.66 (7.58)	216.84 (5.43)	–24.53	–0.20
Observations	222	281	503		

Notes: In this table, we present the characteristics of those who took up either of the asset finance contracts, compared to those who were assigned to a treatment contract but did not take the product. Standard errors are clustered at the individual level. All flow variables are for the last month, and all currency values are in US dollars equivalent based on the prevailing exchange rate during implementation of the projects (US\$-Re of approximately 105). Normalized differences are computed as the difference in means divided by the square root of half of the sum of the variances. (In online Appendix Table A.2, we provide a similar comparison but comparing our full experimental sample with the broader population of just under 30,000 first-time borrowers.)

Asset Choice and Usage.—In this section, we provide further details on the assets chosen by microenterprise owners and how they used them, as well as their understanding of, and satisfaction with, the financing contract that they received.

Online Appendix Figure A.1 illustrates the types of assets funded. The most popular assets selected were rickshaws (33 percent), followed by sewing machines (14 percent), cameras (10 percent), and manufacturing or welding machines

with column 2 of panel B of online Appendix Table A.4, which reports from the survey-based data that outstanding loans in the control group rose from \$33.90 at baseline to \$81.32 by the 6-month mark, a difference of \$47, which is quite close to the aforementioned value inferred from Akhuwat's administrative data. This is reassuring in terms of the accuracy of our data, more generally. Finally, the numbers in panel B of online Appendix Table A.4 also reveal that we are not seeing any significant crowding in or crowding out of borrowing; there are a few people who borrow from other MFIs and family/friends, but after winsorizing the survey data, the differences between total borrowing from Akhuwat and total borrowing from all sources are very small.

TABLE 3—TAKE-UP AND BORROWING

	Any loan (1)	Any loan (2)	Cash loan (3)	Cash loan (4)	Asset loan (5)	Asset loan (6)	Fixed- repayment (7)	Flexible- repayment (8)	Total borrowing (9)	Total borrowing (10)
Assignment	0.48 (0.030)		−0.08 (0.023)		0.56 (0.022)				821.42 (36.947)	
Assignment: Fixed		0.44 (0.038)		−0.09 (0.024)		0.53 (0.031)	0.53 (0.031)			748.87 (50.440)
Assignment: Flexible		0.52 (0.037)		−0.07 (0.025)		0.59 (0.031)	0.09 (0.018)	0.50 (0.032)		897.21 (52.714)
Control mean	0.13	0.13	0.13	0.13	0.00	0.00	0.00	0.00	40.46	40.46
Observations	757	757	757	757	757	757	757	757	757	757

Notes: We report take-up indicators and borrowing amounts for any type of loan (cash or asset-based) from all participants within the first three months of them entering the experiment, using administrative data from the MFI. In online Appendix Table A.4, we conduct a similar exercise without restricting the time period to be the first three months of the experiment (i.e., using administrative data on borrowing throughout the project). *Assignment* refers to assignment to either of the two asset finance contracts (fixed- or flexible-repayment). In columns 1 and 2, the dependent variable equals one if participants took up any new loan; in columns 3 and 4, the dependent variable is a dummy for taking up any cash loan; and in columns 5 and 6, it is take-up of an asset-based loan. In column 7, the dependent variable is a dummy for take-up of the fixed-repayment contract, and in column 8 it is a dummy for take-up of the flexible-repayment contract. In columns 9 and 10, the dependent variable is the total borrowing amount, combining both loan types, in US dollars.

(7 percent). Other popular assets included manufacturing machines, food production machines, computers, photocopiers, and printers. Choice of asset often mapped in a very intuitive way onto baseline business sector. For example, 88 percent of people who bought cameras were working in photography or entertainment, 87 percent of those who purchased sewing machines were in tailoring and its related sectors, 80 percent of those who purchased a food-related machine were working in the food business, and approximately two-thirds of those who purchased manufacturing or welding machines came from that sector. In other sectors, there appeared to be a little more diversification at play with asset choice but still within similar sectors. For example, while half of those who purchased rickshaws were already in the transportation sector, many others were from sectors for which a transportation asset could conceivably function as a complementary business asset (for example, food and drink businesses that might be using the asset for deliveries, as well as retailers in the garment business). As we describe in Section II, our results suggest a large and persistent expansion in the fixed capital stock for treatment clients but no large expansion in the number of business employees. As such, our asset financing appears to have induced a profitable change in the “production technology” for businesses with a large expansion in fixed assets, and in some cases a diversification in the mix of fixed assets. In Section IIB, we also explore whether our treatments increase the rate of sectoral switching; we find little evidence for this. We also confirm, using mediation analysis, that sectoral switching explains very little of our estimated positive effects on profits.

We also asked detailed questions in all follow-up surveys to understand how clients actually used the asset. Respondents report frequent usage: on average, six days per week, and eight hours per day. Ninety-six percent of respondents reported that the asset was regularly used for the business. In terms of *who* was using the asset, 84 percent of the time it was the microenterprise owner themselves, 16 percent of the time

it was business employees, and 7 percent of the time it was some other household member. The numbers are almost exactly the same at all follow-up waves, indicating that—at least in the two years of our project—there is no evidence of severe deterioration in the assets and their usability (something supported by photographs of the assets, taken by our enumerators). Further, as discussed in Appendix Section O, we directly measured respondents' perceptions of depreciation rates using an incentivized task; we estimate this at 5 percent per quarter.

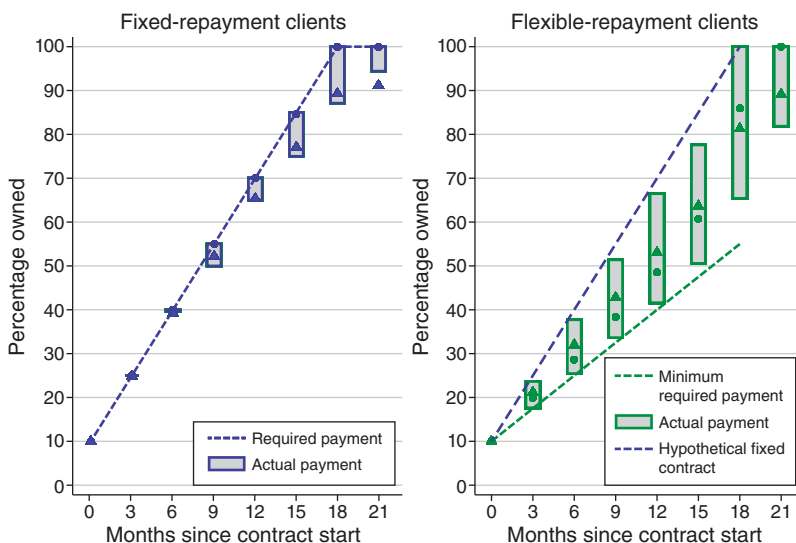
E. Repayment Patterns for the Asset Finance Contracts

All repayments were made in person at a branch. In Figure 1, we illustrate the repayment patterns. Panel A illustrates the trend in actual asset ownership per quarter over the duration of the contract, compared to what was required. The left-hand graph of panel A presents data for clients under the fixed-repayment contract, and the right-hand graph of panel A presents data for clients under the flexible-repayment contract. The mean and median ownership shares are represented by the triangles and circles, respectively, and the interquartile range is represented by the gray-filled bar. The dotted lines illustrate (i) the ownership share required by the fixed-repayment contract (blue) and (ii) the minimum ownership share required by the flexible-repayment contract (green). (We also add a blue dotted line to the right-hand graph of panel A to illustrate what the flexible-repayment clients would have been required to pay under the fixed contract.)

Panel A shows that, from an administrative perspective, the contracts performed well. For clients in the fixed-repayment contract, ownership shares are very close to those formally required (though, as one would expect, loan officers tolerated some occasional repayment delays, particularly toward the end of the 18-month period). The solid blue circle shows that the median client was up to date with required payments. Clients in the flexible-repayment contract generally paid substantially more than the minimum required; at the 18-month mark, the average ownership share for clients under the flexible-repayment contract was 80 percent. While the original agreement was that the asset would be sold in the market and proceeds disbursed in proportion to the ownership shares, in practice—since many clients had repurchased a large share of their asset—by the end of the contract, the MFI allowed them a few extra months to fully purchase the asset (which many clients successfully did).

Panel B of Figure 1 explores in further detail the variation in absolute monthly payment among flexible-contract clients; as in panel A, the green dotted line represents what clients were actually required to pay under the flexible contract, and the blue dotted line illustrates what they hypothetically would have been required to pay under the fixed contract (5 percent of the ownership share each month: twice the required amount under the actual flexible contract). A nontrivial proportion of clients decided to pay monthly amounts that were close to what would have been required under the fixed contract, while some did use the flexibility allowed. There is significant month-to-month variation in repayments made under the flexible contract, mostly lying in between what entrepreneurs were required to pay and what the equivalent payment would have been under the fixed contract. This is consistent with the results of Battaglia, Gulesci, and Madestam (2023), who find that the grace

Panel A. Microenterprise asset ownership share



Panel B. Payments under the flexible contract

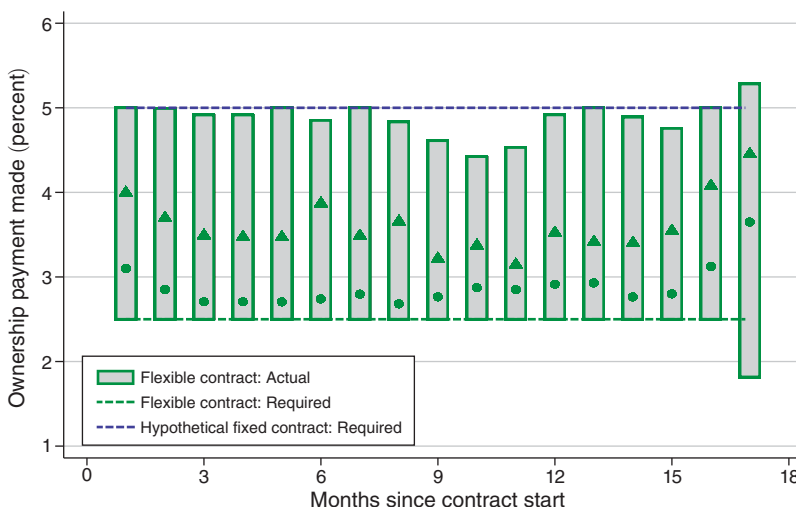


FIGURE 1. ADMINISTRATIVE DATA ON CONTRACT REPAYMENTS

Notes: Panel A illustrates the trend in actual asset ownership over the life of the contract, compared to what was obligated under each contract. The left-hand-side graph of panel A presents data for clients under the fixed-repayment contract, and the right-hand-side graph of panel A presents data for clients under the flexible-repayment contract. The mean and median ownership shares are represented by the triangles and circles, respectively, and the interquartile range is represented by the gray-filled bar. The dotted lines illustrate (i) the ownership share required by the fixed-repayment contract (blue) and (ii) the minimum ownership share required by the flexible-repayment contract (green). In the right-hand-side figure of panel A, the blue dotted line illustrates what the flexible-repayment clients hypothetically would have been required to pay under the fixed contract. Panel B explores in further detail the variation in absolute monthly payment among flexible-contract clients; the green dotted line represents what clients were actually required to pay under the flexible contract, and the blue dotted line illustrates what they hypothetically would have been required to pay under the fixed contract. The mean and median ownership shares are represented by the triangles and circles, respectively, with the interquartile range represented by the gray-filled bar.

periods they offered were used across the loan cycle and sometimes not used at all. (In online Appendix Table A.50, we explore the relationship between usage of the flexible repayment option and shocks faced by the microenterprise; there, we also test how the repayment response varies with baseline risk preferences and volatility of business income.)

The MFI experienced relatively few defaults (fewer than 4 percent of clients), with no significant difference in default between the fixed and flexible contracts. For defaulting clients, the assets were repossessed and sold in the market, as agreed in the original contract. The MFI reported to us that asset repossession and sales were conducted in a straightforward manner in almost all cases, with no reports of clients running away with assets or disputing the contractual terms. At the two-year mark, we included in our survey a module just for those who took the asset, to explore their experience with the product. Ninety percent of clients stated that they understood how the contract worked (specifically, how ownership and rental payments were calculated). Reported understanding was not significantly different across the two contracts. Sixty-eight percent of clients stated that the contract helped them to grow their business (with 22 percent strongly agreeing with that statement).

II. Treatment Effects

In this section, we show the average treatment effects of our two interventions. In doing so, we follow our preanalysis plan (available at www.socialscience-registry.org/trials/3886); we note explicitly in a few places where, to understand mechanisms, we run estimations that were not prespecified. Throughout this analysis, our results follow an intent-to-treat (ITT) specification. We report equivalent local average treatment effect (LATE) estimations in online Appendix Section H.

A. Pooled Results

Our primary analysis pools our two treatment arms. Specifically, we denote T_i as a dummy for whether the respondent was assigned either to treatment 1 or to treatment 2, and we use an ANCOVA specification with strata dummies:

$$(1) \quad y_{it} = \beta_0 + \beta_1 \cdot T_i + \beta_2 \cdot y_{i0} + \phi_{s_i} + \varepsilon_{it}.$$

In doing so, we pool observations from follow-up surveys conducted 3 months, 6 months, 12 months, 18 months, and 24 months after the time of treatment; we cluster errors at the individual level. In each regression table, we report estimated average treatment effects ($\hat{\beta}_1$, in equation (1)), standard errors, p -values, and sharpened q -values (Benjamini, Krieger, and Yekutieli 2006).

The Business.—We begin, in Table 4, by testing effects on key business outcomes for the primary business in the household. We find large and significant effects across a range of key outcomes. Specifically, treated respondents are, on average, 9 percentage points more likely to be running a business (compared to 80 percent of the control

TABLE 4—OVERALL BUSINESS OUTCOMES

	Runs a business (1)	Number of businesses (2)	Business total assets (3)	Business revenue (4)	Business profits (5)	Business employees (6)
Assignment	0.09 (0.02) [0.00] {0.00}	0.10 (0.02) [0.00] {0.00}	401.22 (89.94) [0.00] {0.00}	1.82 (39.65) [0.96] {0.47}	26.93 (9.93) [0.01] {0.01}	0.04 (0.06) [0.54] {0.28}
Control mean (follow-up)	0.80	0.82	1,003.34	689.65	249.31	0.56
Observations	3,608	3,608	3,608	3,608	3,608	3,608

Notes: We report the *intent-to-treat* estimates of the combined treatment on primary outcomes, obtained by least-squares estimation. Below each coefficient, we report a standard error in parentheses, a *p*-value in brackets, and a *q*-value in curly braces. Standard errors allow for clustering at the level of the individual. *q*-values are obtained using the sharpened procedure of Benjamini, Krieger, and Yekutieli (2006).

TABLE 5—BUSINESS ASSETS

	Total fixed assets (1)	Current assets: Cash (2)	Current assets: Accounts receivable (3)	Current assets: Inventory (4)
Assignment	438.05 (67.15) [0.00] {0.00}	2.68 (1.77) [0.13] {0.25}	−0.59 (1.47) [0.69] {0.53}	−29.76 (34.53) [0.39] {0.36}
Control mean (follow-up)	660.19	31.38	9.93	250.77
Observations	3,608	3,608	3,608	3,608

Notes: We report the *intent-to-treat* estimates of the combined treatment on primary outcomes, obtained by least-squares estimation. Below each coefficient, we report a standard error in parentheses, a *p*-value in brackets, and a *q*-value in curly braces. Standard errors allow for clustering at the level of the individual. *q*-values are obtained using the sharpened procedure of Benjamini, Krieger, and Yekutieli (2006).

group).¹⁶ Average business assets are larger by 40 percent of the control group mean (on average, an increase of about \$401 compared to \$1,003 in the control group).¹⁷ This generates an increase in profits of about 11 percent, an average increase of about \$27 on a control group mean of about \$249. We find no effect on employment; this is unsurprising, given the traditional difficulty of encouraging microenterprises to hire workers (see, in particular, De Mel, McKenzie, and Woodruff 2019).

In Table 5, we disaggregate this capital effect into its constituent parts: fixed assets, cash, accounts receivable, and inventory. Our results are stark and

¹⁶In online Appendix Table A.5, we test the effect of treatment on wage employment. Consistent with this result on self-employment, we find that treated respondents are, on average, 7 percentage points less likely to be working in wage employment; as a result, treated respondents work fewer wage jobs, fewer wage hours, and earn less wage income. For analogous results, see Breza and Kinnan (2021).

¹⁷“Business assets” refers to assets *in* the business, regardless of whether the business fully owns them; for example, treated respondents who accepted the contract would report the full market value of the asset, even if they only partially owned it at the time of the follow-up survey. This follows standard accounting practice for capital leases as assets on the balance sheet (provided that there is a transfer of ownership or the option of ownership transfer at the end of the term) as per the Generally Accepted Accounting Principles (US GAAP) and the International Financial Reporting Standards (IFRS), most of which Pakistan has adopted.

unsurprising: all of the effect on total assets is driven by the effect on fixed assets. The magnitude of the increase in fixed assets is intuitive, once we account for reasonable rates of depreciation (discussed shortly).

We find no effect on business revenue—despite finding a significant effect on profits. In part, this difference may reflect the inherent noisiness of measuring microenterprise revenue as opposed to profits (see, for example, De Mel, McKenzie, and Woodruff 2009); it is possible, given the standard errors on business revenue, that the increase in revenue is actually larger than the increase in profits that we observe. In online Appendix Table A.6 we test treatment effects on business costs.¹⁸ We find that the treatment caused a large and significant reduction in business costs—in particular, a reduction of 17 percent in expenditure on raw materials. That is, our profit result is primarily explained by a reduction in business expenses rather than an increase in revenues.¹⁹

Finally, we test for treatment effects on management practices in the microenterprise. To do this, we administered a modified version of the questions used by McKenzie and Woodruff (2016). In online Appendix Table A.8, we find a large and significant effect on management practices concerning inventory purchasing and management. It is possible that some part of this impact might be “mechanical” (for example, a larger asset might require more sophisticated management of inputs). However, the positive effect on purchasing and control of inventories is driven by all three components of that measure. Namely, treated respondents are more likely to attempt to negotiate with suppliers on the price of raw materials, more likely to compare prices from alternative suppliers, and they run out of inventories less frequently. We also rule out the possibility that the improvement in management practices is a mechanical result of encouraging some respondents to switch into running rickshaws: in online Appendix Table A.33, we show that only about 1 percent of this estimated ATE is mediated through the switch into running rickshaws. Further, we also find some evidence of better marketing practices. Note that our two treatments did not include any assistance with management of the asset nor any training on—for example—market access or general business management practices. This finding of better inventory management is also consistent with the earlier results that the increase in overall profits is primarily driven by a reduction in business expenditure on raw materials.

The Household.—Our hire-purchase contracts clearly improved the performance of the microenterprise—but what are the consequences of this for household welfare? To answer this question, we test effects of our treatment on household income, expenditure, and savings; we show results in Table 6. In column 1, we find a large and significant treatment effect on total household monthly income, which increases by about 9 percent relative to the control group. (This effect is driven solely by

¹⁸The analysis in this table was not prespecified; we have conducted these regressions in order to shed further light on our results on business revenues and business profits.

¹⁹Online Appendix Table A.6 also allows us to rule out the possibility that our finding of greater profits is mechanically driven by microenterprise owners receiving a new asset and subsequently reducing their previous expenditure on asset rental and/or old asset repairs; although there is a significant negative effect of the treatment on machine rent expenses, the magnitude is small (a \$3 decrease per month), and machine repair costs actually marginally increase (by \$1 per month).

TABLE 6—HOUSEHOLD OUTCOMES

	Total household income	Household consumption expenditure	Total household savings	Household assets
	(1)	(2)	(3)	(4)
Assignment	31.47 (12.66) [0.01] {0.02}	12.95 (3.37) [0.00] {0.00}	16.44 (19.16) [0.39] {0.24}	20.33 (14.03) [0.15] {0.11}
Control mean (follow-up)	357.35	220.40	113.03	681.79
Observations	3,608	3,608	3,608	1,410

Notes: We report the *intent-to-treat* estimates of the combined treatment on primary outcomes, obtained by least-squares estimation. Below each coefficient, we report a standard error in parentheses, a *p*-value in brackets, and a *q*-value in curly braces. Standard errors allow for clustering at the level of the individual. *q*-values are obtained using the sharpened procedure of Benjamini, Krieger, and Yekutieli (2006).

the treatment effect on business profits; in separate regressions, we find no significant effect on other sources of household income.) In column 2, we find a significant effect on total monthly household consumption expenditure (an increase of 6 percent relative to the control group). Although the coefficient is positive and relatively large, we find no statistically significant effect on total household savings (column 3).²⁰

When we disaggregate the increase in household consumption (Table 7), we find a striking result: our treatment caused a large and significant effect on households' expenditure on schooling. Specifically, we find an increase of 26 percent on the control group mean, from about \$22 per month to about \$28. Table 8 provides a further disaggregation of household educational expenditure into its constituent subcategories (we collected these measures in the 24-month endline survey, prompted by having found significant effects on schooling expenditure in the previous follow-up rounds). In panel A, we show significantly greater overall schooling expenditure on both girls and boys (an increase of 25 percent and 17 percent, respectively, relative to the control mean), with both effects highly significant even after multiple hypothesis corrections. Panel B reveals that the increased spending on children's education is evident across all the measured subcategories of spending: the treatment effects on school fees; spending on books, stationery, and other materials; spending on school meals as well as school transportation costs. Coefficients are again highly significant even after correction for multiple hypothesis testing across expenditure categories. The effects are positive and significant for both girls and boys, although the estimated effects are generally greater for girls.

Following these results, we went back to the field to get a better insight into the mechanisms driving these effects on education. We called clients from the take-up group who had experienced the biggest increase in schooling expenditure between baseline and follow-up; we did this in order to obtain descriptive evidence on the primary motivations of those individuals driving the education treatment effects. We

²⁰In our preanalysis plan we had included total household borrowing in this table of outcomes, but we now report a much more detailed analysis of total borrowing in Table 3 and online Appendix Table A.4.

TABLE 7—DISAGGREGATING HOUSEHOLD CONSUMPTION

	Food (1)	Clothing (2)	Bills (3)	Schooling (4)	Health (5)	Special occasions (6)	Household items (7)
Assignment	2.61 (0.90) [0.00] {0.01}	1.54 (1.82) [0.40] {0.42}	0.67 (0.31) [0.03] {0.05}	5.70 (1.30) [0.00] {0.00}	0.09 (0.30) [0.78] {0.61}	0.04 (0.29) [0.89] {0.61}	1.91 (1.39) [0.17] {0.20}
Control mean (follow-up)	52.80	34.71	24.54	22.05	2.24	7.30	67.54
Observations	3,608	3,608	3,608	3,608	3,608	3,608	3,608

Notes: We report the *intent-to-treat* estimates of the combined treatment on primary outcomes, obtained by least-squares estimation. Below each coefficient, we report a standard error in parentheses, a *p*-value in brackets, and a *q*-value in curly braces. Standard errors allow for clustering at the level of the individual. *q*-values are obtained using the sharpened procedure of Benjamini, Krieger, and Yekutieli (2006).

used a structured survey that allowed for long-text responses. We received a high response rate (89 percent), with 100 surveys completed. Around half of respondents mentioned that they increased the number of hours that they sent children to school (for example, extra tuition centers or summer schools). Over a third mentioned that they switched to a different school. Some of the most popular reasons given by respondents for why they believed that educational expenditure will pay off in terms of their children’s future earnings was that it would enable their children to (i) find a good job, (ii) become financially independent, (iii) achieve financial stability, (iv) “break the poverty cycle,” and (v) improve health. Many respondents also mentioned confidence, self-respect, and social recognition.

Our education results are consistent with the recent work of Agte et al. (2021), following up the earlier work of Field et al. (2013) on flexible-repayment microcredit (whose original intervention, like ours, successfully led to greater microenterprise investment, assets, and income). Agte et al. (2021) find that economic benefits persisted and spilled over to the next generation. Specifically, treatment households spend more on private secondary schooling and after-school tutoring on children—and, subsequently, children who were school-aged at baseline were more likely to attend college. Our result is also consistent with Jack et al. (2023), who, like us, offer asset-based microfinance (in their case, rainwater-harvest tanks in an agricultural setting) and who find positive treatment effects on girls’ school enrollment.

In Figure 3, we show two empirical CDFs: one for total consumption and one for consumption on schooling. Each graph shows a clear separation of CDFs: a general shift of the distribution to the right. Table 7 also shows a significant increase in expenditure on food, of about 5 percent of the control group mean (from about \$53 to \$56 per month). This result stands in clear contrast to previous research on microfinance; it suggests that financing the purchase of a productive asset may generate sustained improvements in household welfare as well as improving microenterprise performance, specifically in terms of households’ investment in their children’s human capital.

Finally, in online Appendix Table A.7, we test for effects on respondents’ attitudes toward saving (including respondents’ reports of savings problems, making of unnecessary purchases, feeling pressure to share, and similar outcomes). We find no effect on any of these measures.

TABLE 8—IMPACTS ON CHILDREN’S EDUCATION

	In school: Girls (1)	In school: Boys (2)	Expenditure: Girls (3)	Expenditure: Boys (4)				
<i>Panel A. Extensive margin and total schooling expenditures</i>								
Assignment	0.04 (0.03) [0.16] {0.12}	0.04 (0.03) [0.24] {0.14}	4.83 (1.31) [0.00] {0.00}	3.13 (1.32) [0.02] {0.03}				
Control mean (follow-up)	0.83	0.85	19.55	18.81				
Observations	549	487	549	487				
	School fees: Girls (1)	School fees: Boys (2)	Books & materials: Girls (3)	Books & materials: Boys (4)	Food expenditure: Girls (5)	Food expenditure: Boys (6)	Transport costs: Girls (7)	Transport costs: Boys (8)
<i>Panel B. Schooling expenditure categories</i>								
Assignment	1.53 (0.80) [0.06] {0.03}	0.89 (0.77) [0.25] {0.07}	0.81 (0.25) [0.00] {0.00}	0.50 (0.26) [0.06] {0.03}	0.99 (0.34) [0.00] {0.00}	1.23 (0.35) [0.00] {0.00}	1.46 (0.39) [0.00] {0.00}	0.67 (0.35) [0.06] {0.03}
Control mean (follow-up)	11.43	10.34	3.25	3.56	2.69	2.97	2.83	2.77
Observations	553	493	553	493	553	493	553	493

Notes: We report the ITT estimates of the combined treatment on outcomes related to children’s education, obtained by least-squares estimation. Below each coefficient, we report a standard error in parentheses, a p -value in brackets, and a q -value in curly braces. Standard errors allow for clustering at the level of the individual. q -values are obtained using the sharpened procedure of Benjamini, Krieger, and Yekutieli (2006).

B. Disaggregation

Separating Treatment 1 and Treatment 2.—In online Appendix Section I, we repeat our earlier analysis, splitting by whether respondents were assigned to treatment 1 or treatment 2. We find no robust differences in average outcomes between these treatments. In online Appendix Table A.40, we show that the difference between treatment 1 and treatment 2 matters when considering heterogeneity in risk aversion; however, there is no robust difference in average effects of the two treatments across the sample as a whole.

Separating by Survey Wave.—In online Appendix Section J, we reproduce all of our main previous results, disaggregating by each survey wave. We present estimates individually for follow-up surveys at the 3-month, 6-month, 12-month, 18-month, and 24-month points. Our results are very stable across waves: we see no large differences in coefficients for any of our prespecified outcome variables across time. In particular, we note that the majority of our estimated effects remain large even at the 24-month follow-up; this is itself an important aspect of our results (that is, the fact that our control group does not catch up over time) and one that we explore shortly with our structural model. We note that there is suggestive evidence that the fixed capital effect declines slightly over time; in online Appendix Section K, we show that this is consistent with reasonable assumptions about depreciation.

As online Appendix Section J illustrates, there is suggestive evidence of an increasing treatment effect on business profits over time. In online Appendix Section L,

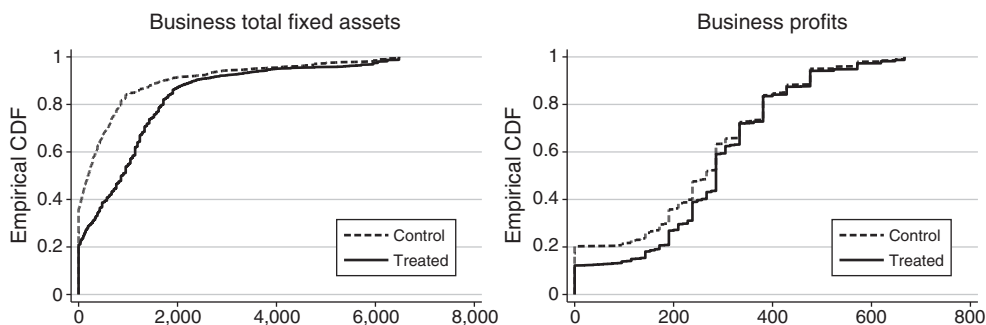


FIGURE 2. EMPIRICAL CDFS: BUSINESS TOTAL FIXED ASSETS AND BUSINESS PROFITS

Note: In the above empirical CDFs for business total fixed assets and business profits, we pool all follow-up survey waves.

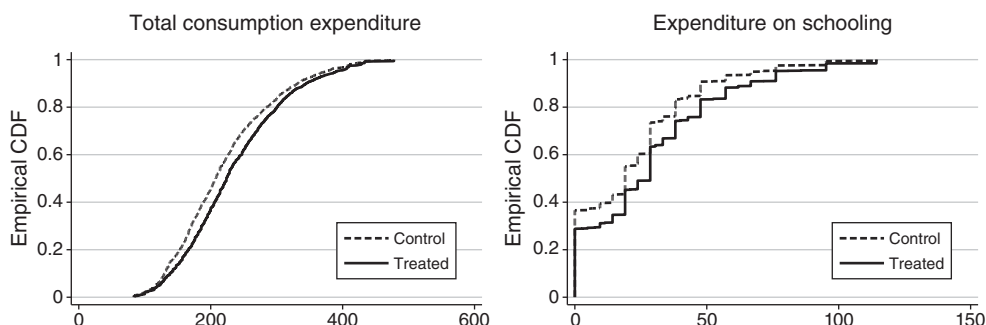


FIGURE 3. EMPIRICAL CDFS: CONSUMPTION

Note: The left panel presents the empirical CDF for total household consumption expenditure, and the right panel shows the empirical CDF for the subcomponent representing total household expenditure on schooling (including spending on school fees, books and other materials, food, and transportation).

we explore whether this effect might be related to a change in business activities; we conclude that, overall, there is little in the way of sectoral switching. In online Appendix Section M, we explore whether sectoral switching might explain some of our estimated positive effects on profits. We use the method of Acharya, Blackwell, and Sen (2016) to calculate the average controlled direct effect, using as a mediator a dummy variable for whether the respondent runs a rickshaw (the main asset in the most popular business sector in our sample, transportation). We find that this mediator explains about 30 percent of the estimated ATE on raw materials and on bills; however, the mediator explains only about 8 percent of the estimated effect on profits.

C. Robustness

We test robustness both to outliers and to endogenous attrition. Online Appendix Section N considers outliers: there, we take the main treatment effects of interest from our previous analysis and subject them to increasing degrees of winsorization. Specifically, we report winsorizing (top and bottom) at 2.5 percent (used for our

original analysis), 1 percent, 5 percent, and 10 percent. Our results remain remarkably stable across specifications, including their statistical significance. This is entirely consistent with the empirical CDFs (see Figures 2 and 3), which show that our treatment effects apply across the distribution (rather than, for example, only appearing in the tails).

Attrition is very low for this sample: the overall attrition rate is just under 4 percent.²¹ Further, attrition is uncorrelated with treatment: when we estimate equation (1) using a dummy for attrition as the outcome variable (and, of course, omitting the ANCOVA term), we obtain a treatment effect of just 0.7 percentage points, with $p = 0.55$. For these reasons, we conclude that our analysis is robust to concerns about endogenous attrition.

III. Structural Analysis: Adjustment Costs, Wealth Dynamics, and Contractual Design

We now specify and calibrate a dynamic structural model. We do this for three related reasons. First, the model helps us to understand how a large capital injection, financed through a hire-purchase contract, can generate large and sustained improvements in household wealth and income. Second, in doing so, the model helps us to characterize microenterprise dynamics and to understand how our treatments affected those dynamics. Third, we then extend the model to allow for counterfactual scenarios.

Our general approach here is broadly similar to two seminal structural models of microfinance: those of Kaboski and Townsend (2011) and of Banerjee et al. (2021). Our model builds on these earlier contributions by incorporating explicitly an asset-based financing product of the form implemented in our experiment. Specifically, our structural estimates describe a world in which there is a low return to holding cash or other liquid assets. This means that households choose to hold only minimal liquid assets over time. Credit-constrained households are therefore unwilling to accumulate sufficient liquid wealth to overcome substantial nonconvex capital adjustment costs (costs that are driven, for example, by the indivisibility of fixed assets)—even though, if purchased, such assets would have a high productive value to the household microenterprise. This kind of juxtaposition—between high-return illiquid assets and low-return liquid assets—has been noted in several recent empirical contexts. In particular, it is central to Kaboski and Townsend’s (2011) structural analysis of the “Million Baht” program in Thailand.²² The same juxtaposition has also recently been applied to household behavior in heterogeneous-agent new Keynesian macroeconomic models, where it is described as generating a “wealthy hand-to-mouth” phenomenon (see, in particular, Kaplan and Violante 2014 and the discussion in Kaplan, Violante, and Weidner 2014). Our model implies that house-

²¹ As one would expect, this rate increases with the time since the baseline survey. However, the wave-by-wave attrition rates remain low for all waves: for the 3-month follow-up, attrition is 2.4 percent; for the 6-month follow-up, it is 2.6 percent; for the 12-month follow-up, it is 4.6 percent; for the 24-month follow-up, it is 5.9 percent; and for the 36-month follow-up, it is 7.8 percent.

²² Kaboski and Townsend (2011) provide for lumpy investments with complete irreversibility; they allow such investments to have a return “higher than the interest rate on liquid savings, r , and sufficiently high to induce investment for households with high enough liquidity” (p. 1373).

hold wealth levels are likely to be highly persistent and that there are profitable and persistent gains from microfinance products that provide large capital injections.

A. Model Specification

The No-Contract Case: Our basic model describes a credit-constrained household that runs a microenterprise and optimizes on an infinite horizon in discrete time:

$$(2) \quad V_n(k_t, f_t, \varepsilon_t, \psi_t) = \max_{k_{t+1}, f_{t+1}} E_{(\varepsilon_{t+1}, \psi_{t+1}) | (\varepsilon_t, \psi_t)} \left[\frac{c_t^{1-1/\gamma}}{1 - 1/\gamma} + \beta \cdot V_n(k_{t+1}, f_{t+1}, \varepsilon_{t+1}, \psi_{t+1}) \right],$$

subject to

$$(3) \quad c_t = (1 - \tau) \cdot \exp(\mu + \varepsilon_t) \cdot k_t^\alpha - \Delta k_t - \delta \cdot k_t - s_t - a_t > 0;$$

$$(4) \quad s_t = f_{t+1} - (1 + r) \cdot f_t;$$

$$(5) \quad \varepsilon_{t+1} | \varepsilon_t \sim \mathcal{N}(\rho \cdot \varepsilon_t, \sigma^2).$$

Here, the state space comprises fixed capital (k_t), a liquid financial asset (f_t), a productivity shock (ε_t), and a dummy for whether the household has an investment opportunity (ψ_t). The Bellman equation (2) is formed by assuming that the household maximizes the expected discounted future utility of consumption. Equation (3) explains that the household obtains income through the microenterprise (where we assume a value added production function that is Cobb-Douglas in fixed capital, having total factor productivity of $\exp(\mu + \varepsilon_t)$). We allow for an ad valorem kinship tax on microenterprise income (τ); this is intended primarily to reflect community “sharing norms,” by which the respondent household is expected to contribute to poorer households in the extended family and broader community (Jakiela and Ozier 2016; Squires 2021). We define Δk_t as the change in fixed capital ($\Delta k_t \equiv k_{t+1} - k_t$); capital depreciation is $\delta \cdot k_t$. We use a_t for capital adjustment costs, defined shortly (equation (7)). Equation (4) is a standard savings equation, in which r represents the real return on saving. As discussed shortly, we allow a slightly negative real interest rate, r , on the assumption that savings are largely held in cash and often without effective savings devices (Dupas and Robinson 2013). Equation (5) allows both for productivity shocks and for persistent entrepreneurial ability.

To this basic setup, we add four important constraints, which we view as important realities of running a microenterprise in a low-income country. First, like Banerjee et al. (2021), we assume that—absent formal microfinance contracts—households are credit constrained: $f_t \geq 0$. Second, fixed capital is lumpy: a household cannot, for example, buy or sell a rickshaw one wheel at a time. The assumption of lumpiness is a common feature of several key models of microfinance (see, for example, Besley, Coate, and Lounry 1993; Kaboski and Townsend 2011; Field et al. 2013; and Banerjee, Karlan, and Zinman 2015). This assumption reflects the reality that respondents do not have access to liquid rental markets for fixed capital (nor, indeed, to sophisticated norms or contractual forms to allow for time-sharing in fixed capital

usage, as in Bassi et al. 2022). Empirically, it reflects the observation that a large number of enterprises in our data do not adjust their fixed capital from one period to the next. It also reflects the observation that many enterprises that make such capital adjustments do so by making a discrete switch from one line of business into another.²³ Formally, we follow Field et al. (2013) by modeling such lumpiness through imposing a minimum investment size (κ); we view this assumption as a useful way of capturing nonconvex adjustment costs more generally. Third, like Kaboski and Townsend (2011), we allow investment opportunities to be stochastic; we do this by allowing ψ_t to be drawn independently each period from a Bernoulli distribution having parameter $\omega < 1$. Formally, we require that Δk_t belongs to one of three line segments:

$$(6) \Delta k_t \in \begin{cases} \{[-(1 - \delta) \cdot k_t, -\kappa], [-\delta \cdot k_t, 0]\}, & \text{if } \psi_t = 0; \\ \left\{ \underbrace{[-(1 - \delta) \cdot k_t, -\kappa]}_{\text{sell}}, \underbrace{[-\delta \cdot k_t, 0]}_{\text{repair}}, \underbrace{[\kappa - \delta \cdot k_t, \infty)}_{\text{buy}} \right\}, & \text{if } \psi_t = 1. \end{cases}$$

Here, the segment $[-(1 - \delta) \cdot k_t, -\kappa]$ corresponds to a situation where the household is selling fixed capital; we require a minimum sale of size κ .²⁴ The segment $[\kappa - \delta \cdot k_t, \infty)$ corresponds to a situation where the household purchases fixed capital; here, we require a minimum purchase of value κ . Together, when $\kappa > 0$, these two line segments imply a nonconvex adjustment cost in capital: what Bloom (2009) describes as “a central region of inaction” (see also Caballero and Engel 1999 and Cooper and Haltiwanger 2006). Note that this investment segment is unavailable to the household when $\psi_t = 0$. To this, we add a small intermediate segment for replacement investment, $[-\delta \cdot k_t, 0]$, which corresponds to a situation in which the household neither buys nor sells fixed capital but chooses to repair some share of the depreciation.²⁵

Fourth, we assume that fixed capital is partially irreversible—in the sense that sales of fixed capital incur a proportionate markdown in capital value, $\phi \in [0, 1]$ (as in Ramey and Shapiro 2001 and Cooper and Haltiwanger 2006):

$$(7) \quad a_t = \begin{cases} -\phi \cdot (\Delta k_t + \delta \cdot k_t), & \text{if } \Delta k_t + \delta \cdot k_t < 0 \\ 0, & \text{otherwise.} \end{cases}$$

²³Previous empirical work shows that business start-up costs for urban microenterprises can be substantial. In particular, Fafchamps and Quinn (2017) study aspiring entrepreneurs in Ethiopia, Tanzania, and Zambia and show large effects on business start-up from cash grants of \$1,000. Klinger and Schündeln (2011) show large effects for grants between \$6,000 and \$15,000; McKenzie (2017) shows large effects from grants with a median size of \$57,000.

²⁴Of course, the household cannot sell more fixed capital than it owns. Note that, for households having $k_t < \kappa \cdot (1 - \delta)^{-1}$, this first segment is a null set; in that case, asset sales are not possible. In online Appendix Section O.9, we consider an alternative specification in which the minimum transaction size applies to capital expansions but not to capital reductions; we show that this alternative specification barely changes the model behavior in this context.

²⁵Thus, for example, a household can sell a rickshaw, or buy a rickshaw, or add a new coat of paint to repair general wear and tear on the rickshaw. But no amount of new paint will turn one rickshaw into two. For this reason, note that the upper bound of the first segment corresponds to a situation in which the household sells fixed capital but pays the depreciation on the existing capital; the lower bound of the third segment corresponds to a situation where the household buys fixed capital but allows the existing capital to depreciate.

Our model therefore combines both nonconvexities (in the form of the capital adjustment costs) and financial market frictions (in the form of household credit constraints). As many authors have noted—including, recently, Ghatak (2015); Banerjee et al. (2021); and Balboni et al. (2022)—this combination opens the possibility that the effects of large capital shocks are highly persistent.²⁶

Introducing Microfinance: This basic setup can be adjusted to allow for microfinance—first in the form of a standard unconditional loan of \$475 and then in the form of a \$1,500 asset-finance contract that mimics our fixed-repayment hire-purchase agreement. To model the standard loan, we introduce a new state variable, x_t ; this is an integer count of the household's point in a loan cycle (such that $x_t = 0$ reflects the start of the cycle, $x_t = X$ is the final period of repayment, and x_t increments by one each period). We then write a new value function, incorporating this state variable; we also assume that the household is lent some lump-sum F to be repaid in X periods (with zero interest); we do this by relaxing the lower bound on the financial asset, such that $f_t \geq -F + x_t \cdot F/X$. Alternatively, to model the fixed-repayment asset financing contract described earlier, we retain the assumption $f_t \geq 0$ and adjust equation (4) to account for the repayment structure required by that contract. We explain these amendments in detail in online Appendix Section O.

B. Solution and Calibration

We solve the model in two steps. First, we solve for V_n (the no-contract case); this is a stationary infinite-horizon problem (by equation (2)) and can be solved by standard numerical contraction. Second, with the solution to V_n in hand, we solve for the two separate microfinance cases using backward induction over the fixed number of repayment periods. We then obtain relevant moments (described shortly) by simulating forwards through the model solution, starting from the observed baseline joint distributions of (k_t, f_t) (and implementing the asset-finance contract for the treatment group). Our model does not endogenize contract take-up; instead, we simulate take-up using the observed empirical proportions who respectively took each contract type (independent of the simulated point in the state space).

We use several different methods to calibrate the model parameters; these are summarized in Table 9 and described in detail in online Appendix Section O. We obtain the production function parameters (μ and α) and the productivity persistence and variance (ρ and σ^2) by using a quasi-differenced GMM estimator (Blundell and Bond 2000; Cooper and Haltiwanger 2006). We rely on an incentivized belief-elicitation exercise to obtain values for δ and ϕ . We back out τ by an accounting exercise using baseline household control group averages for consumption, business profits, asset sales, and net saving. To pin down the probability of having an investment opportunity, we use the take-up rate under treatment 1: that is, $\omega = 0.52$. We assume a quarterly discount factor of $\beta = 0.9$, and we use $\gamma = 0.35$ as the intertemporal

²⁶ As Kaboski and Townsend (2011, pp. 1360–61) put it, “given the lumpiness of projects, small amounts of credit are relatively unlikely to change investment decisions on large projects.”

TABLE 9—CALIBRATED STRUCTURAL PARAMETERS

Parameter	Description	Value	Source
μ	Mean of log productivity	5.93	Panel GMM
ρ	Quarterly autocorrelation of productivity	0.62	Panel GMM
σ	Standard deviation of productivity	0.30	Panel GMM
α	Curvature of production	0.16	Panel GMM
r	Quarterly real return on saving	−0.0125	Implied by inflation
δ	Quarterly depreciation rate	0.05	Incentivized measure
ϕ	Partial irreversibility cost	0.25	Incentivized measure
τ	Ad valorem sharing tax	0.15	Baseline accounting
ω	Probability of investment opportunity	0.52	Take-up under treatment 1
γ	Intertemporal elasticity of substitution	0.35	Assumed
β	Quarterly discount factor	0.90	Assumed
κ	Minimum investment size	1,500	See online Appendix Figure A.7

Notes: This table reports a series of structural parameter values used for our calibration exercise. “Panel GMM” refers to a quasi-differenced GMM panel estimator; “incentivized measure” refers to a series of incentivized lab-in-field games conducted at baseline; “baseline accounting” refers to an accounting exercise using baseline data. We provide further detail in online Appendix Section O.

elasticity.²⁷ Finally, we choose a negative quarterly real interest rate, $r = -1.25\%$; this would approximate a setting with an annual inflation rate of 5 percent and a zero nominal interest on saving. (We show shortly that our estimates are robust to a very wide range of plausible assumptions about r .) With these parameter values in hand, we then search over a grid of possible values for κ , in order to understand the importance of nonconvex adjustment costs. We evaluate these different values using an Indirect Inference loss function, in which we target treatment effects on fixed capital, value added, and household consumption; we target these effects at the 3-month, 6-month, 12-month, 18-month, and 24-month follow-ups.

C. Results: Household Behavior under Nonconvex Adjustment Costs

We find that our treatment effects are rationalized much more effectively by a model with large nonconvex capital adjustment costs than a more standard model with no such costs; we illustrate this in online Appendix Figure A.7, where we show how the Indirect Inference loss function varies with κ . In this context, we view the purpose of this structural exercise *not* as identifying a single κ that should be taken very literally as a minimum investment size; as noted earlier, κ serves here as a stylized device to capture nonconvex adjustment costs through capital lumpiness, and our results here indicate that such costs are economically large and meaningful. Indeed, our estimate of large nonconvex capital adjustment costs is consistent with treated respondents’ decisions to purchase very valuable assets; as noted earlier, the median asset purchase was \$1,666, and approximately one-third of respondents chose the maximum financing amount possible. (Based on online Appendix Figure A.7, we choose $\kappa = 1,500$.)

The model with large nonconvex costs fits the observed data well, in several respects. First, the model replicates closely the pattern of *targeted moments*. We

²⁷ We chose $\gamma = 0.35$ to match the estimate for India in Ogaki, Ostry, and Reinhart (1996). We chose $\beta = 0.9$ to reflect the stylized observation—from low-stakes incentivized baseline games—that the respondent pool has relatively high impatience over cash.

show this in online Appendix Figure A.8; this figure shows the real treatment effects (for fixed capital, value added, and consumption, at all follow-up waves), and we superimpose simulated treatment effects under both the $\kappa = 1,500$ and $\kappa = 0$ model variants. Our preferred model replicates large and persistent treatment effects on both fixed capital and enterprise value added.

It is worth noting that—even under $\kappa = 1,500$ —the structural treatment effects on fixed capital decline noticeably faster than the experimental estimates suggest (though the structural effects remain within the experimental confidence intervals). This suggests that the actual capital adjustment costs are more intricate than our (very stylized) representation allows; in particular, there may be an important role for permanent heterogeneity, which our model ignores in the interests of simplicity and tractability.²⁸ Similarly, the fit on consumption is close to the observed treatment effect, though nonetheless smaller in absolute terms. As we discussed earlier, our reported consumption treatment effects include spending on children's education—which is, in many respects, more of an investment in human capital than the kind of consumption that the model describes. In contrast, the treatment effects cannot be replicated by the $\kappa = 0$ version of the model; in that version, the control group is able to catch up quickly, both in terms of fixed capital and value added.

Second, the model also replicates well a large number of *untargeted moments*. Specifically, we compare model predictions to data for fixed capital (both in levels and in first differences), for value added (in levels and in differences), for household consumption (in levels and in differences), and for financial assets (in levels); we do this both for control and treatment groups, at the 3-month, 6-month, 12-month, 18-month, and 24-month marks, and we map the twenty-fifth, fiftieth, and seventy-fifth percentiles. In online Appendix Figure A.9, we compare real and simulated moments and show that the fit for our preferred model is remarkably close to the 45-degree line (in online Appendix Section O, we provide graphical comparisons for each of these moments, separately for the control and treatment groups). Again, the model fit is much better under $\kappa = 1,500$ than $\kappa = 0$. In particular, under $\kappa = 0$, the model predicts substantially more capital accumulation, both in control and treatment groups, than is actually observed.

More generally, our preferred model framework rationalizes three key features of our data—and, indeed, data from many microenterprise studies in the literature. *First*, most microenterprises in our sample make little or no adjustment to their fixed capital stock over time; it is *not* the case, for example, that households steadily build their wealth by multiple incremental investments in fixed capital. Indeed, our data on period-to-period changes in fixed assets reveal that the median 6-monthly change for the control group was 0 (as it was at the seventy-fifth percentile); even at the ninetieth percentile, the change in capital is only \$300, and we only observe increases in total fixed capital of \$1,000 or more in 7 percent of our follow-up data for the control group.²⁹ *Second*, notwithstanding this fact, the marginal product of

²⁸In particular, we note that in online Appendix Figure A.10, the deterioration in the capital fit is driven by the model predicting that, in the control group, some firms will accumulate substantial capital through realization of high productivity draws. At substantial cost to tractability and interpretability, one could generalize the current model by allowing permanent heterogeneity in μ and in κ .

²⁹Similarly, Balboni et al. (2022) argue that the opportunity for individuals to significantly increase their productive assets would not have arisen without the program that they study: only 5.6 percent of their control group

fixed capital in the microenterprise is high (De Mel, McKenzie, and Woodruff 2008; Fafchamps et al. 2014). Specifically, we estimate $\alpha = 0.16$; this is similar to other microenterprise production function estimates in other contexts (Janes, Koelle, and Quinn 2022; Atkin, Khandelwal, and Osman 2017), and, for the firm sizes in our dataset, implies a high marginal return to fixed capital. For example, for a firm having fixed capital of \$500 and with $\varepsilon_t = 0$, this implies a marginal product of capital of $\alpha \cdot \exp(\mu) \cdot 500^{\alpha-1}$: a return of about 33 percent. *Third*, most households in our data hold minimal wealth in cash or other liquid assets (Dupas and Robinson 2013).

Online Appendix Figure A.12 shows policy functions, both for fixed capital k (in the left panel) and financial capital f (in the right panel). (We illustrate the policy functions for the no-credit case in order to highlight the underlying tension between the choice of the two different forms of capital.) Specifically, the figure illustrates the stark implications for household capital accumulation in our preferred model: given both the opportunity and the cash, households would willingly invest in fixed capital. However, large nonconvex adjustment costs mean that these high returns to capital lie beyond the reach of most households; instead, those same households rationally consume their available cash.³⁰ Rather, interventions that facilitate the acquisition of productive indivisible assets can have highly persistent impacts, by shifting the household to a new point within what is essentially a stable *range* of states. Online Appendix Figure A.13 illustrates this stability using a phase diagram in (k, f) space. In short, our model implies that a microfinance intervention offering a relatively small lump-sum payment will not generate transformational change to the household's circumstances. The household will rationally spend such a payment to increase consumption in the short run; in our model, such a payment will not suffice for investment in fixed capital and will prove too costly to be held in cash. As a consequence, microfinance interventions that allow the household to accumulate a larger lump sum (such as the intervention described in this paper and the "grace period" innovation of Field et al. 2013) can generate persistent improvements in both wealth and welfare.

D. Counterfactual Scenarios

We now use the structural model to run two sets of counterfactual analyses. We view each set of counterfactual analyses as serving two related purposes: (i) the analysis allows us to explore further the mechanisms driving the structural results, and (ii) the analysis helps us to think about policy alternatives.

Variation in the Real Interest Rate.—Our first counterfactual analysis considers variation in the real interest rate. In our main specification, we used a quarterly real interest rate of $r = -1.25\%$. As equation (4) showed, this is the rate earned on the liquid financial asset. We can, therefore, think of counterfactual increases in r literally as representing improvements in the interest rate on savings; more generally, this can describe changes in the attractiveness of other outside investment

experience a change of log assets of the same magnitude as their average asset injection.

³⁰ We noted earlier that the general behavior described by our model mirrors that of the "wealthy hand-to-mouth" model of Kaplan and Violante (2014). The right panel of online Appendix Figure A.12 shows that our model replicates one of the key implications of that earlier literature: namely, a very high marginal propensity to consume out of shocks to cash (observed in our model at least for households with relatively low levels of physical capital).

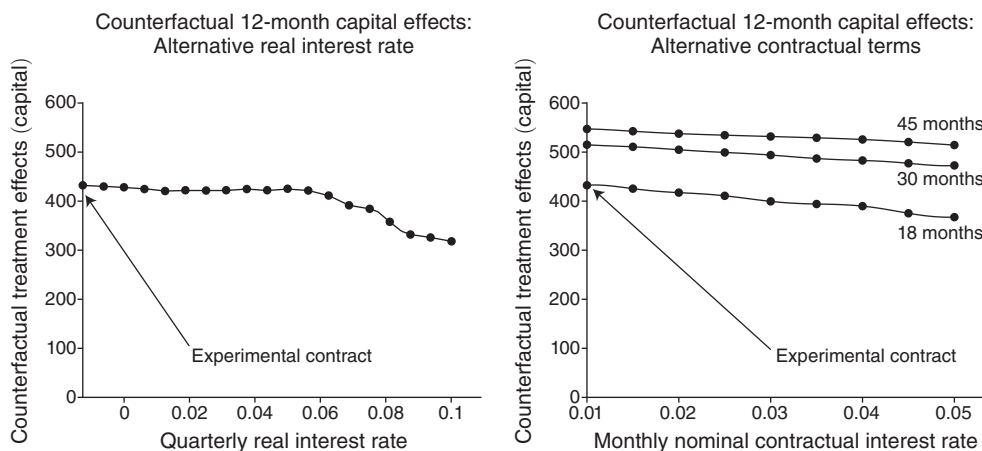


FIGURE 4. COUNTERFACTUAL STRUCTURAL ANALYSIS

Notes: These panels show 12-month counterfactual treatment effects on fixed capital. In both cases, we rerun our main model specification using the parameters in Table 9 and $\kappa = 1,500$. In the left panel, we consider alternative values for the real interest rate, r . In the right panel, we consider alternative contractual interest rates and contractual durations; in each case, we keep the 10 percent deposit (amortizing the 90 percent loan over the period of the contract).

options. We implement this counterfactual by rerunning our main model specification (using the parameters in Table 9 and $\kappa = 1,500$) across a grid of values for r : from $r = -1.25\%$ to $r = 10\%$ (that is, an annual interest rate over 40 percent). We show counterfactual 12-month treatment effects on fixed capital in the left panel of Figure 4 (with counterfactual treatment effects on value added and consumption in online Appendix Section O.7).

As one would expect, the general slope with respect to the real interest rate is negative: as the value of holding cash increases, the household optimally chooses less fixed capital. However, what is striking about the left panel of Figure 4 is how gentle is the decline: the increase in real interest rate only has an appreciable impact on the capital treatment effect once the quarterly real interest reaches about 8 percent (and, even then, it remains within the 95 percent confidence interval of the actual experimental treatment effect). This result, that the structural predictions are robust to a very wide range of plausible values for the real interest rate, makes strong intuitive sense given the key mechanism described in Section IIIC. In a model without adjustment costs, household behavior would be highly sensitive to the relative return to fixed capital and the real interest rate (indeed, this comparison would determine firm size). Under large nonconvex adjustment costs, the marginal product of fixed capital can already far exceed the real interest rate (as noted in Section IIIC); so it makes intuitive sense that even relatively large changes in that real interest rate have only limited implications for model behavior.

More generally, this counterfactual highlights the central relevance of capital adjustment costs for a range of alternative policy initiatives. In different ways, various policies seek to improve the real return on liquid savings in low-income settings: for example, through improved savings technologies, business training programs, through insulating entrepreneurs from sharing pressures, and so on. Our model results, though

necessarily stylized, suggest that such policies are unlikely to generate persistent gains if the marginal propensity to consume from liquid savings is high.

Variation in the Contractual Terms.—Our second counterfactual exercise considers plausible variations in the contractual terms. Recall that—as Table 1 illustrates—our contract requires a 10 percent deposit, with the remaining 90 percent amortized across the 18 months. To this obligation, we add a 1 percent nominal monthly interest payment. Under our counterfactual contracts, we keep the 10 percent deposit requirement and vary both (i) the contract duration (in each case, amortizing the 90 percent loan over the entire period of the contract) and (ii) the nominal contractual interest rate. To run these counterfactuals, we rerun our main model specification (again, using the parameters in Table 9 and $\kappa = 1,500$), adjusting the repayment terms and the duration of the contract (and, therefore, of the finite-horizon backward induction). As we do so, we hold fixed both take-up rates and default rates (neither of which our model endogenizes); this simplification assists the clarity and transparency of the exercise (and is an assumption that we relax in the following section when considering benefit-cost ratios).

The right panel of Figure 4 illustrates the counterfactual 12-month treatment effects on fixed capital (the equivalent graphs for value added and consumption are provided in online Appendix Section O.8). As one would expect, clients are wealthier when the interest rate is lower and when the contract duration is longer: in both cases, the MFI is charging less in return for the same capital injection. More interesting, and more relevant for policy, is the magnitude of the change. First, the simulations indicate that gains remain substantial for monthly nominal interest rates up to and including 5 percent (though, as noted, one might expect take-up to drop substantially under such high rates). Second, the simulations indicate that gains are likely to be about 25 percent larger if the contract were offered over 30 months rather than 18 months (with smaller gains anticipated when moving from 30 months to 45 months). We explore these counterfactuals further in the following section, when considering benefit-cost ratios under higher interest rates.

IV. Benefit-Cost Analysis

We now explore benefit-cost ratios and internal rates of return (IRR). In doing so, we build upon the methodology of Banerjee et al. (2015) and Bandiera et al. (2017). To do this, we use actual implementation costs and estimated treatment effects; we then follow Alfonsi et al. (2020) in using our model-based counterfactuals to calculate benefit-cost ratios and IRR under alternative contract structures that may be relevant for potential future scale-up.

In Table 10, we present the key elements of the analysis (all in US dollars). Costs are comprised of (i) capital disbursed for the initial purchase of the assets for take-up clients, subtracted from total capital recovered (i.e., building in all nonpayment of contractual obligations); (ii) staff salaries; and (iii) all other implementation costs. Total costs are then compounded to the two-year mark using a social discount rate of 10 percent, which is on the conservative side of the range of social discount rates recommended by the World Bank (Lopez 2008). We provide two estimates for each cost: (i) a “lower bound,” where we take the total amount that was reported to us as

TABLE 10—BENEFIT-COST ANALYSIS: KEY ELEMENTS

	Project total	Cost lower bound: Per treatment client (<i>N</i> = 503)	Cost upper bound: Per take-up client (<i>N</i> = 281)
<i>Costs</i>			
Capital disbursed for initial purchase of assets	388,571	773	1,383
Total capital recovered from clients (including defaults)	−385,524	−766	−1,372
Total capital disbursed minus capital recovered (discounted to year 0)	38,095	76	136
Staff salaries (calculated as if all incurred at start of year 0)	30,076	60	107
Other implementation costs (calculated as if all incurred at start of year 0)	1,810	4	6
Total cost (calculated as of year 0)	69,981	139	249
Total costs compounded to year 2 at 10 percent social discount rate		168	301
<i>Benefits</i>			
		Average (ITT)	
Year 1 business profit treatment effect		440	
Year 2 business profit treatment effect		396	
Year 2 business assets treatment effect		329	
Total benefits at year 2		1,165	
<i>Total business profits year 3 onward, assuming benefits last:</i>			
	1 year	360	
	2 years	687	
	3 years	985	
	5 years	1,501	
	10 years	2,433	
	15 years	3,012	
	20 years	3,371	

Notes: We provide the key elements for our analysis of benefit-cost ratios and internal rates of return (IRR), the final output of which is presented in Figure 5. All values are in USD. Costs are reported to us by the MFI. For benefits, we use treatment effects from ITT regressions (excluding individuals assigned to the flexible-repayment contract, for consistency with the structural model), as well as an estimate of future benefits beyond the period of the project (the net present value of all future benefits from year 3 onward, by taking the year-2 ITT on business profits as the annual value of future benefits, and showing the result under various assumptions for persistence).

project costs by the MFI and divide it by the 503 individuals *assigned* to treatment (giving an average total cost of \$168 per client), and (ii) an “upper bound,” where the total cost is divided by the smaller number of 281 who actually took up the contract (giving an average cost of \$301).

For benefits, we use treatment effects from ITT regressions (excluding individuals assigned to the flexible-repayment contract, for consistency with the structural model), as well as an estimate of future benefits beyond the period of the project. For benefits during the project period, we sum the treatment effects on business profits in the first and second year after asset disbursement, as well as the year-2 treatment effect on business assets. This gives a total benefit of \$1,165 during the project period. To this, we add the estimated net present value of all future benefits from year 3 onward, by taking the year-2 ITT on business profits as the annual value of future benefits. We conduct our analysis of benefit-cost ratios and IRR under various assumptions for the persistence of benefits, ranging from 0 years of future benefit to persistence for 20 years (implying an NPV of future benefits of \$3,371).

We present the results of the analysis graphically in Figure 5, both as a benefit-cost ratio and an IRR. In each graph, the horizontal axis represents the number of years the benefits are assumed to persist, and we show results using both the lower- and

upper-bound cost estimates. Panel A represents the analysis using the actual implementation costs and the estimated treatment effects (described in Table 10), with the top graph illustrating the benefit-cost ratio and the bottom graph illustrating the IRR. We estimate very high benefit-cost ratios, even under very conservative assumptions. For example, using the upper-bound cost estimates, we find a benefit-cost ratio of 3.9 even when assuming zero persistence of effects after the second year of implementation. This rises to a ratio of 8.8 when assuming 5 years of benefits, and 11.9 when benefits persist for 10 years. The IRR is 109 percent even when we assume 0 years of persistence, and the IRR converges to about 140 percent when assuming 5 years or more of persistence.

In panel B, we take the model-based estimates for treatment effects at year 1 and at year 2, under the actual implemented contract cost (1 percent per month); note that this imposes slightly conservative figures for treatment effects relative to the experimental point estimates. Nevertheless, estimated benefit-cost ratios remain very high. Using the upper-bound cost estimates, we find a benefit-cost ratio of 2.7 even when assuming zero persistence of effects, which rises to 5.8 when assuming 5 years of benefits, and 7.7 with 10 years of persistence. The IRR is 51 percent when we assume 0 years of persistence, and the IRR converges to approximately 88 percent when assuming 5 years or more of persistence.

Finally, in panel C of Figure 5, we use model-based estimates for treatment effects using a counterfactual cost of 2 percent per month. This reduces the estimated treatment effects, to which we also add a conservative estimate of a doubling of losses to the MFI from additional defaults under the more expensive contract (see online Appendix Table A.38 for the extent to which we increase costs and reduce benefits). Even with these conservative assumptions, we continue to find high rates of return. We find a benefit-cost ratio of 1.7 if we assume zero persistence of effects, rising to a ratio of 3.7 when assuming 5 years of benefits, and 4.9 when benefits persist for 10 years. The IRR is 9 percent when we assume 0 years of persistence, rising to 31 percent with only 1 year of persistent benefits, 48 percent with 3 years of persistent benefits, and converging to about 55 percent when assuming 5 years or more of persistence.

V. Conclusions

In this paper, we test the effects of a hire-purchase contract, which facilitates large capital injections for experienced microfinance clients. We find large and significant gains in microenterprise assets and profits, and in turn, increases in household income and consumption. We show that our results can be rationalized by a structural model that allows for large nonconvex adjustment costs in fixed capital. We then use this model to consider counterfactual scenarios, including potential variations on some key contractual terms. In this concluding section, we now discuss the scope for scaling such a product in different settings.

Our IRR analysis indicates that the product was highly cost-effective, under a range of plausible assumptions about the persistence of treatment effects. In practical terms, there are several related reasons why our large treatment effects generate a large IRR. Most fundamentally, our capital injection was provided as a loan—rather than as a grant—which very substantially reduces the net cost to the provider. Put

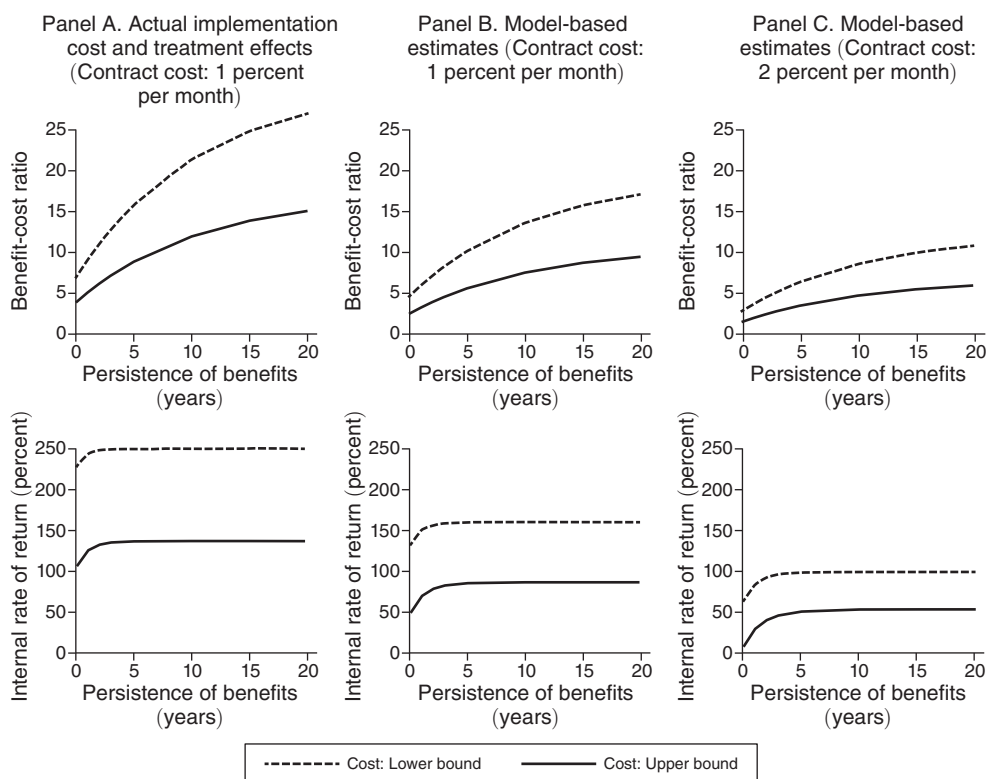


FIGURE 5. BENEFIT-COST RATIOS AND INTERNAL RATES OF RETURN

Notes: In this figure, we present benefit-cost ratios and internal rates of return. In panel A, we use the actual estimated treatment effects and project implementation costs. In panel B, model-based estimates are used, based on a 1 percent per month contract cost (which is the actual contract implemented in the experiment). In panel C, model-based counterfactual treatment effect estimates are used, based on a contract cost of 2 percent per month, which affects both the benefits (impacts of the product on business and household income) as well as the costs (a higher rate of return for the MFI from the higher price of the contract, to which we also add a higher risk of default—twice the actual loss faced by the MFI during the project).

differently, our partner MFI recovered almost all of its initial outlay, allowing it to redeploy those funds to other borrowers. Further, we observed a high repayment rate, though we also note that our IRR would remain very large even if (for example) the interest rate were set to more traditional market rates and if the default rate were to double. Finally, we note that all of the financing was used to purchase productive assets: we did not include any spending on (for example) training on asset use, or business mentoring. Such additions would have increased substantially the cost and the complexity of the program; we speculate that these add-ons were unnecessary in this context given that we selected experienced clients who had successfully repaid previous business loans (and who had identified a specific asset that would benefit their business).

Nonetheless, when considering the scaling up of any intervention like ours, credible enforcement mechanisms are obviously important. In our setting, the MFI used a combination of quite traditional approaches. First, the MFI screened clients to be graduated borrowers, who had showed a willingness and an ability to repay at least one previous loan, and who had specifically identified a profitable investment

opportunity through the acquisition of a fixed asset. Second, like most lenders, the MFI relies on dynamic incentives as part of its ongoing relationship with borrowers, including a clear and credible threat to refuse future lending in the case of a client's strategic default. Third, as our results showed, the capital investments themselves were highly profitable on average: providing clients with the ability to repay and perhaps further strengthening clients' sense of obligation to do so. Further, the MFI also used traditional in-person methods for asset procurement and for repayment collection. In different contexts, recent technological changes might substantially reduce implementation costs and improve credit risk management and enforcement mechanisms and could form part of any potential scaling up; indeed, many MFIs have dramatically accelerated their digital transformation process in the wake of COVID-19.

More generally, there are a range of related asset-based financing options that deserve further attention from both research and policy angles. In this paper, we use a lease-based structure, but the substance of the contract is very similar to other asset-financing contracts used for SMEs in high-income countries. In an extensive survey of asset financing models across the globe, Kumaraswamy, Mattern, and Hernandez (2020) recommend that lease-based contracts are used for financing higher-value, "productive" (income-generating) assets. Within the category of lease-based models, the authors recommend "rent-to-own" structures (very similar to our hire-purchase agreement) since the collateralization possibilities improve credit risk management. There is also increasing interest and work by the World Bank on collateral registries for movable assets (Love, Pería, and Singh 2016).³¹ Several start-ups are now experimenting with such rent-to-own models for productive asset financing; for example, financing for vehicles such as motorcycle taxis, solar water pumps, heaters and purifiers, farm equipment, and other business assets.

To understand our respondents' recommendations for potential scale-up, we added a series of questions to our final interview round—in which we asked clients their opinions regarding the "optimal contract structure" if the MFI were to offer this product to others in future. First, we asked about the optimal initial payment. The average was 10 percent (with minimal variation): the figure we had implemented. Second, we asked clients what the maximum level of financing should be. The average response was approximately double the amount that we permitted in the current project: \$3,500 (with the seventy-fifth percentile at \$5,000). Third, we asked clients their views on optimal contract duration: the average response was 33 months, with the seventy-fifth percentile at 36 months. Finally, we were interested to know how clients would have used the financing if it had been offered in the form of cash rather than a direct injection of a fixed asset (a question for which we allowed multiple categories of answer). Ninety-five percent of clients reported that they would have used at least some of the funds to purchase a fixed asset, 42 percent said they would have used some of the funds on working capital, and 8 percent said that they would have used some funds for construction or repairs to their business premises. Among those reporting that they would have purchased a fixed asset, clients reported, on average,

³¹ See also Campello and Larrain (2016) for evidence that expanding the menu of assets that may be used as loan collateral (to include movable assets) can lead to an increase in firms' access to credit, investment, and productivity.

that 80 percent of the funds would have been used for that asset. This highlights the importance of financing that provides fixed assets as well as complementary working capital for graduated borrowers.

Together, our results show that large asset-based microfinance contracts have the potential to stimulate microenterprise growth among graduated borrowers. Of course, large asset-based financing is not for everyone: for many microenterprise owners, smaller and more standard loans may be more appropriate given their risk profile and investment requirements. As the microfinance sector matures, with a growing focus on “financial inclusion,” it is increasingly important that MFIs are able to expand the suite of products they offer in their portfolio, to allow clients to obtain products that are well suited to their individual requirements. Given their sharia-compliant contractual form, asset-based contracts are likely to have particular appeal to Muslim entrepreneurs—who face disproportionately high levels of financial exclusion (Karim, Tarazi, and Reille 2008; El-Gamal et al. 2014; Demirguc-Kunt, Klapper, and Randall 2014; Karlan, Osman, and Shammout 2021). There is also no reason for these advantages to be limited to Islamic contexts; indeed, asset-based financing is an important source of credit for small businesses around the world and a form of contract that could readily be extended to many microenterprises.

REFERENCES

- Acharya, Avidit, Matthew Blackwell, and Maya Sen. 2016. “Explaining Causal Findings Without Bias: Detecting and Assessing Direct Effects.” *American Political Science Review* 110 (3): 512–29.
- Agte, Patrick, Arielle Bernhardt, Erica M. Field, Rohini Pande, and Natalia Rigol. 2021. “Investing in the Next Generation: The Long-Run Educational Impacts of a Liquidity Shock.” Unpublished.
- Ahmed, Malik Laique. 2019. *Pakistan Microfinance Review 2018*. Islamabad, Pakistan: Pakistan Microfinance Network.
- Ahmed, Malik Laique. 2020a. *Microwatch (Issue 53): A Quarterly Update on Microfinance Outreach in Pakistan*. Islamabad, Pakistan: Pakistan Microfinance Network.
- Ahmed, Malik Laique. 2020b. *Microwatch (Issue 54): A Quarterly Update on Microfinance Outreach in Pakistan*. Islamabad, Pakistan: Pakistan Microfinance Network.
- Alfonsi, Livia, Oriana Bandiera, Vittorio Bassi, Robin Burgess, Imran Rasul, Munshi Sulaiman, and Anna Vitali. 2020. “Tackling Youth Unemployment: Evidence from a Labor Market Experiment in Uganda.” *Econometrica* 88 (6): 2369–2414.
- Angelucci, Manuela, Dean Karlan, and Jonathan Zinman. 2015. “Microcredit Impacts: Evidence from a Randomized Microcredit Program Placement Experiment by Compartamos Banco.” *American Economic Journal: Applied Economics* 7 (1): 151–82.
- Athey, S., and G. W. Imbens. 2017. “The Econometrics of Randomized Experiments.” In *Handbook of Economic Field Experiments*, Vol. 1, edited by Abhijit Vinayak Banerjee and Esther Duflo, 73–140. Amsterdam: Elsevier.
- Atkin, David, Amit K. Khandelwal, and Adam Osman. 2017. “Exporting and Firm Performance: Evidence from a Randomized Experiment.” *Quarterly Journal of Economics* 132 (2): 551–615.
- Attanasio, Orazio, Britta Augsborg, Ralph De Haas, Emla Fitzsimons, and Heike Harmgart. 2015. “The Impacts of Microfinance: Evidence from Joint-liability Lending in Mongolia.” *American Economic Journal: Applied Economics* 7 (1): 90–122.
- Augsburg, Britta, Ralph De Haas, Heike Harmgart, and Costas Meghir. 2015. “The Impacts of Microcredit: Evidence from Bosnia and Herzegovina.” *American Economic Journal: Applied Economics* 7 (1): 183–203.
- Balboni, Clare, Oriana Bandiera, Robin Burgess, Maitreesh Ghatak, and Anton Heil. 2022. “Why Do People Stay Poor?” *Quarterly Journal of Economics* 137 (2): 785–844.
- Bandiera, Oriana, Robin Burgess, Narayan Das, Selim Gulesci, Imran Rasul, and Munshi Sulaiman. 2017. “Labor Markets and Poverty in Village Economies.” *Quarterly Journal of Economics* 132 (2): 811–70.
- Banerjee, Abhijit, Emily Breza, Esther Duflo, and Cynthia Kinnan. 2021. “Can Microfinance Unlock a Poverty Trap for Some Entrepreneurs?” Unpublished.

- Banerjee, Abhijit, Esther Duflo, Rachel Glennerster, and Cynthia Kinnan.** 2015. "The Miracle of Microfinance? Evidence from a Randomized Evaluation." *American Economic Journal: Applied Economics* 7 (1): 22–53.
- Banerjee, Abhijit, Esther Duflo, Nathanael Goldberg, Dean Karlan, Robert Osei, William Parienté, Jeremy Shapiro, Bram Thuysbaert, and Christopher Udry.** 2015. "A Multifaceted Program Causes Lasting Progress for the Very Poor: Evidence from Six Countries." *Science* 348 (6236): 1260799.
- Banerjee, Abhijit, Dean Karlan, Robert Osei, Hannah Trachtman, and Christopher Udry.** 2022. "Unpacking a Multi-faceted Program to Build Sustainable Income for the Very Poor." *Journal of Development Economics* 155: 102781.
- Banerjee, Abhijit, Dean Karlan, and Jonathan Zinman.** 2015. "Six Randomized Evaluations of Microcredit: Introduction and Further Steps." *American Economic Journal: Applied Economics* 7 (1): 1–21.
- Barboni, Giorgia, and Parul Agarwal.** 2018. "Knowing What's Good for You: Can a Repayment Flexibility Option in Microfinance Contracts Improve Repayment Rates and Business Outcomes?" Unpublished.
- Bari, Faisal ® Kashif Malik ® Muhammad Meki ® Simon Quinn.** 2024. "Replication data for: Asset-Based Microfinance for Microenterprises: Evidence from Pakistan." American Economic Association [Publisher], Inter-university Consortium for Political and Social Research [Distributor]. <https://doi.org/10.3886/E195984V1>.
- Basharat, Ali, and Zeenoor S. Sheikh.** 2019. *The How and the Why of Microfinance Lending Rates*. Islamabad, Pakistan: Pakistan Microfinance Network.
- Bassi, Vittorio, Raffaella Muoio, Tommaso Porzio, Ritwika Sen, and Esau Tugume.** 2022. "Achieving Scale Collectively." *Econometrica* 90 (6): 2937–78.
- Battaglia, Marianna, Selim Gulesci, and Andreas Madestam.** 2023. "Repayment Flexibility and Risk Taking: Experimental Evidence from Credit Contracts." *Review of Economic Studies*. DOI: 10.1093/restud/rdad107.
- Bauchet, Jonathan, and Jonathan Morduch.** 2013. "Is Micro Too Small? Microcredit vs. SME Finance." *World Development* 43: 288–97.
- Benjamini, Yoav, Abba M. Krieger, and Daniel Yekutieli.** 2006. "Adaptive Linear Step-up Procedures that Control the False Discovery Rate." *Biometrika* 93 (3): 491–507.
- Besley, Timothy, Stephen Coate, and Glenn Loury.** 1993. "The Economics of Rotating Savings and Credit Associations." *American Economic Review* 83 (4): 792–810.
- Bloom, Nicholas.** 2009. "The Impact of Uncertainty Shocks." *Econometrica* 77 (3): 623–85.
- Blundell, Richard, and Stephen R. Bond.** 2000. "GMM Estimation with Persistent Panel Data: An Application to Production Functions." *Econometric Reviews* 19 (3): 321–40.
- Breza, Emily, and Cynthia Kinnan.** 2021. "Measuring the Equilibrium Impacts of Credit: Evidence from the Indian Microfinance Crisis." *Quarterly Journal of Economics* 136 (3): 1447–97.
- Bryan, Gharad T., Dean Karlan, and Adam Osman.** 2022. "Big Loans to Small Businesses: Predicting Winners and Losers in an Entrepreneurial Lending Experiment." Unpublished.
- Bursztyn, Leonardo, Stefano Fiorin, Daniel Gottlieb, and Martin Kanz.** 2019. "Moral Incentives in Credit Card Debt Repayment: Evidence from a Field Experiment." *Journal of Political Economy* 127 (4): 1641–83.
- Caballero, Ricardo J., and Eduardo M. R. A. Engel.** 1999. "Explaining Investment Dynamics in US Manufacturing: A Generalized (S, s) Approach." *Econometrica* 67 (4): 783–826.
- Cai, J., M. Meki, S. Quinn, E. Field, C. Kinnan, J. Morduch, and F. Said.** 2021. "Microcredit." *Vox-DevLit* 1 (2): 3.
- Campello, Murillo, and Mauricio Larrain.** 2016. "Enlarging the Contracting Space: Collateral Menus, Access to Credit, and Economic Activity." *Review of Financial Studies* 29 (2): 349–83.
- Carney, Kevin, Michael Kremer, Xinyue Lin, and Gautam Rao.** 2022. "The Endowment Effect and Collateralized Loans." Unpublished.
- Cooper, Russell W., and John C. Haltiwanger.** 2006. "On the Nature of Capital Adjustment Costs." *Review of Economic Studies* 73 (3): 611–33.
- Crépon, Bruno, Florencia Devoto, Esther Duflo, and William Parienté.** 2015. "Estimating the Impact of Microcredit on Those Who Take It Up: Evidence from a Randomized Experiment in Morocco." *American Economic Journal: Applied Economics* 7 (1): 123–50.
- Crépon, Bruno, Mohamed El Komi, and Adam Osman.** Forthcoming. "Is It Who You Are or What You Get? Comparing the Impacts of Loans and Grants for Microenterprise Development." *American Economic Journal: Applied Economics*.
- Cull, Robert, Asli Demirgüç-Kunt, and Jonathan Morduch.** 2018. "The Microfinance Business Model: Enduring Subsidy and Modest Profit." *World Bank Economic Review* 32 (2): 221–44.

- De Mel, Suresh, David McKenzie, and Christopher Woodruff.** 2008. "Returns to Capital in Microenterprises: Evidence from a Field Experiment." *Quarterly Journal of Economics* 123 (4): 1329–72.
- De Mel, Suresh, David J. McKenzie, and Christopher Woodruff.** 2019. "Labor Drops: Experimental Evidence on the Return to Additional Labor in Microenterprises." *American Economic Journal: Applied Economics* 11 (1): 202–35.
- De Mel, Suresh, David J. McKenzie, and Christopher Woodruff.** 2009. "Measuring Microenterprise Profits: Must we ask how the sausage is made?" *Journal of Development Economics* 88 (1): 19–31.
- Demirguc-Kunt, Asli, Leora Klapper, and Douglas Randall.** 2014. "Islamic Finance and Financial Inclusion: Measuring Use of and Demand for Formal Financial Services among Muslim Adults." *Review of Middle East Economics and Finance* 10 (2): 177–218.
- Duflo, Esther.** 2020. "Field Experiments and the Practice of Policy." *American Economic Review* 110 (7): 1952–73.
- Dupas, Pascaline, and Jonathan Robinson.** 2013. "Why Don't the Poor Save More? Evidence from Health Savings Experiments." *American Economic Review* 103 (4): 1138–71.
- El-Gamal, Mahmoud, Mohamed El-Komi, Dean Karlan, and Adam Osman.** 2014. "Bank-insured RoSCA for Microfinance: Experimental Evidence in Poor Egyptian Villages." *Journal of Economic Behavior and Organization* 103: S56–S73.
- Fafchamps, Marcel, David McKenzie, Simon Quinn, and Christopher Woodruff.** 2014. "Microenterprise Growth and the Flypaper Effect: Evidence from a Randomized Experiment in Ghana." *Journal of Development Economics* 106: 211–26.
- Fafchamps, Marcel, and Simon Quinn.** 2017. "Aspire." *Journal of Development Studies* 53 (10): 1615–33.
- Fafchamps, Marcel, and Christopher Woodruff.** 2016. "Identifying Gazelles: Expert Panels vs. Surveys as a means to Identify Firms with Rapid Growth Potential." *World Bank Economic Review* 31 (3): 670–86.
- Fiala, Nathan.** 2018. "Returns to Microcredit, Cash grants and Training for Male and Female Microentrepreneurs in Uganda." *World Development* 105: 189–200.
- Field, Erica, Rohini Pande, John Papp, and Natalia Rigol.** 2013. "Does the Classic Microfinance Model Discourage Entrepreneurship among the Poor? Experimental Evidence from India." *American Economic Review* 103 (6): 2196–2226.
- Gertler, Paul, Brett Green, and Catherine Wolfram.** 2023. "Digital Collateral." Unpublished.
- Ghatak, Maitreesh.** 2015. "Theories of Poverty Traps and Anti-poverty Policies." *World Bank Economic Review* 29 (1): S77–S105.
- Goebel, Roger J.** 1961. "Reconstructing the Roman Law of Real Security." *Tulane Law Review* 36: 29.
- Hussam, Reshmaan, Natalia Rigol, and Benjamin N. Roth.** 2022. "Targeting High Ability Entrepreneurs using Community Information: Mechanism Design in the Field." *American Economic Review* 112 (3): 861–98.
- Imbens, Guido W., and Donald B. Rubin.** 2015. *Causal Inference in Statistics, Social, and Biomedical Sciences*. Cambridge: Cambridge University Press.
- Jack, William, Michael Kremer, Joost de Laat, and Tavneet Suri.** 2023. "Borrowing Requirements, Credit Access, and Adverse Selection: Evidence from Kenya." *Review of Economic Studies* 90 (6): 3153–85.
- Jakiela, Pamela, and Owen Ozier.** 2016. "Does Africa Need a Rotten Kin Theorem? Experimental Evidence from Village Economies." *Review of Economic Studies* 83 (1): 231–68.
- Janes, Laurin, Michael Koelle, and Simon Quinn.** 2022. "Do Capital Grants Improve Microenterprise Productivity?" Unpublished.
- Kaboski, Joseph P., and Robert M. Townsend.** 2011. "A Structural Evaluation of a Large-scale Quasi-experimental Microfinance Initiative." *Econometrica* 79 (5): 1357–1406.
- Kaplan, Greg, and Giovanni L. Violante.** 2014. "A Model of the Consumption Response to Fiscal Stimulus Payments." *Econometrica* 82 (4): 1199–1239.
- Kaplan, Greg, Giovanni L. Violante, and Justin Weidner.** 2014. "The Wealthy Hand-to-Mouth." *Brookings Papers on Economic Activity* 44 (1): 77–138.
- Karim, Nimrah, Michael Tarazi, and Xavier Reille.** 2008. *Islamic Microfinance: An Emerging Market Niche*. Washington, DC: World Bank.
- Karlan, Dean, Adam Osman, and Nour Shammout.** 2021. "Increasing Financial Inclusion in the Muslim World: Evidence from an Islamic Finance Marketing Experiment." *World Bank Economic Review* 35 (2): 376–97.
- Karlan, Dean, and Jonathan Zinman.** 2011. "Microcredit in Theory and Practice: Using Randomized Credit Scoring for Impact Evaluation." *Science* 332 (6035): 1278–84.
- Klinger, Bailey, and Matthias Schündeln.** 2011. "Can Entrepreneurial Activity be Taught? Quasi-experimental Evidence from Central America." *World Development* 39 (9): 1592–1610.

- Kumaraswamy, Sai Krishna, Max Mattern, and Emilio Hernandez.** 2020. "Assets Matter to Poor People: But What Do We Know about Financing Assets?" Unpublished.
- Lensink, Robert, and Erwin Bulte.** 2019. "Can We Improve the Impact of Microfinance? A Survey of the Recent Literature and Potential Avenues for Success." In *The Oxford Handbook of Banking*, edited by Allen N. Berger, Philip Molyneux, and John O. S. Wilson, 404–30.
- Lopez, Humberto.** 2008. "The Social Discount Rate: Estimates for Nine Latin American Countries." Unpublished.
- Love, Inessa, Maria Soledad Martinez Peria, and Sandeep Singh.** 2016. "Collateral Registries for Movable Assets: Does their Introduction Spur Firms' Access to Bank Financing?" *Journal of Financial Services Research* 49 (1): 1–37.
- Mahmud, Mahreen, and Zaki Wahhaj.** 2019. "Charitable Giving or Signalling? Voluntary Contributions by Micro-credit Borrowers in Pakistan." *Journal of Economic Behavior and Organization* 158: 394–415.
- Malik, Kashif, Muhammad Meki, Jonathan Morduch, Timothy Ogden, Simon Quinn, and Farah Said.** 2020. "COVID-19 and the Future of Microfinance: Evidence and Insights from Pakistan." *Oxford Review of Economic Policy* 36 (1): S138–S168.
- McKenzie, David.** 2017. "Identifying and Spurring High-Growth Entrepreneurship: Experimental Evidence from a Business Plan Competition." *American Economic Review* 107 (8): 2278–2307.
- McKenzie, David, and Dario Sansone.** 2019. "Predicting Entrepreneurial Success is Hard: Evidence from a Business Plan Competition in Nigeria." *Journal of Development Economics* 141: 102369.
- McKenzie, David, and Christopher Woodruff.** 2016. "Business Practices in Small Firms in Developing Countries." *Management Science* 63 (9): 2967–81.
- Meager, Rachael.** 2019. "Understanding the Average Impact of Microcredit Expansions: A Bayesian Hierarchical Analysis of Seven Randomized Experiments." *American Economic Journal: Applied Economics* 11 (1): 57–91.
- MIX.** 2019. "Mix market:mfi company metadata." World Bank. <https://datacatalog.worldbank.org/dataset/mix-market>.
- Ogaki, Masao, Jonathan D. Ostry, and Carmen M. Reinhart.** 1996. "Saving Behavior in Low- and Middle-income Developing Countries: A Comparison." *IMF Staff Papers* 43 (1): 38–71.
- Pakistan Bureau of Statistics.** 2017. *Household Integrated Economic Survey 2015–16*. Islamabad, Pakistan: Statistics Division, Government of Pakistan.
- Pitt, Mark M., and Shahidur R. Khandker.** 1998. "The Impact of Group-based Credit Programs on Poor Households in Bangladesh: Does the Gender of Participants Matter?" *Journal of Political Economy* 106 (5): 958–96.
- Ramey, Valerie A., and Matthew D. Shapiro.** 2001. "Displaced Capital: A Study of Aerospace Plant Closings." *Journal of Political Economy* 109 (5): 958–92.
- Roodman, David, and Jonathan Morduch.** 2014. "The Impact of Microcredit on the Poor in Bangladesh: Revisiting the Evidence." *Journal of Development Studies* 50 (4): 583–604.
- Squires, Munir.** 2021. "Kinship Taxation as an Impediment to Growth: Experimental Evidence from Kenyan Microenterprises." Unpublished.
- Tarozzi, Alessandro, Jaikishan Desai, and Kristin Johnson.** 2015. "The Impacts of Microcredit: Evidence from Ethiopia." *American Economic Journal: Applied Economics* 7 (1): 54–89.